

Essentials of Innovation & Entrepreneurship (EIE) - II

Module (M7):

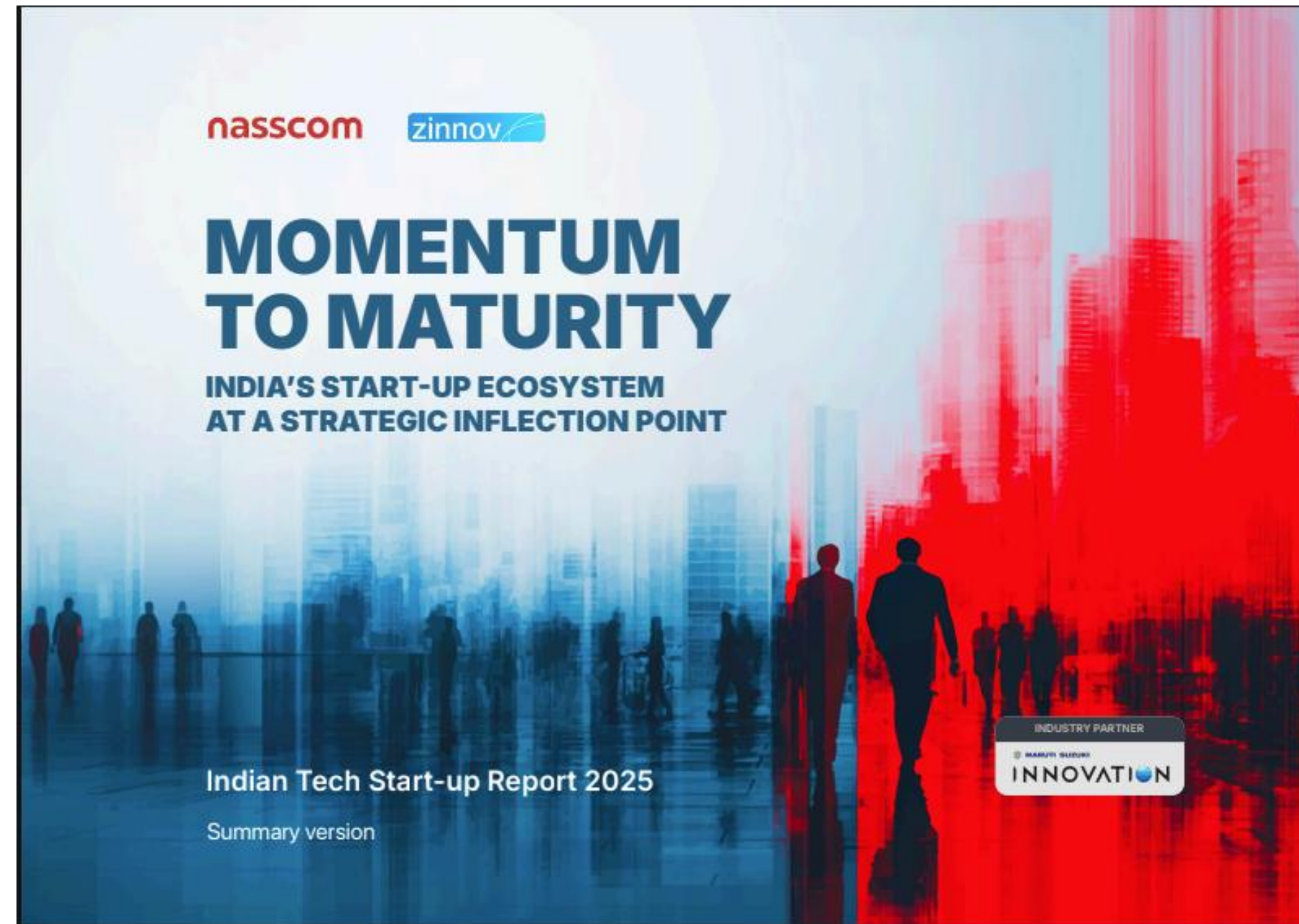
What is (not) Marketing?

Introduction: AI-driven Market Research

EIE2-Icebreaker:

I&E Roundup this Week (Mar 11)

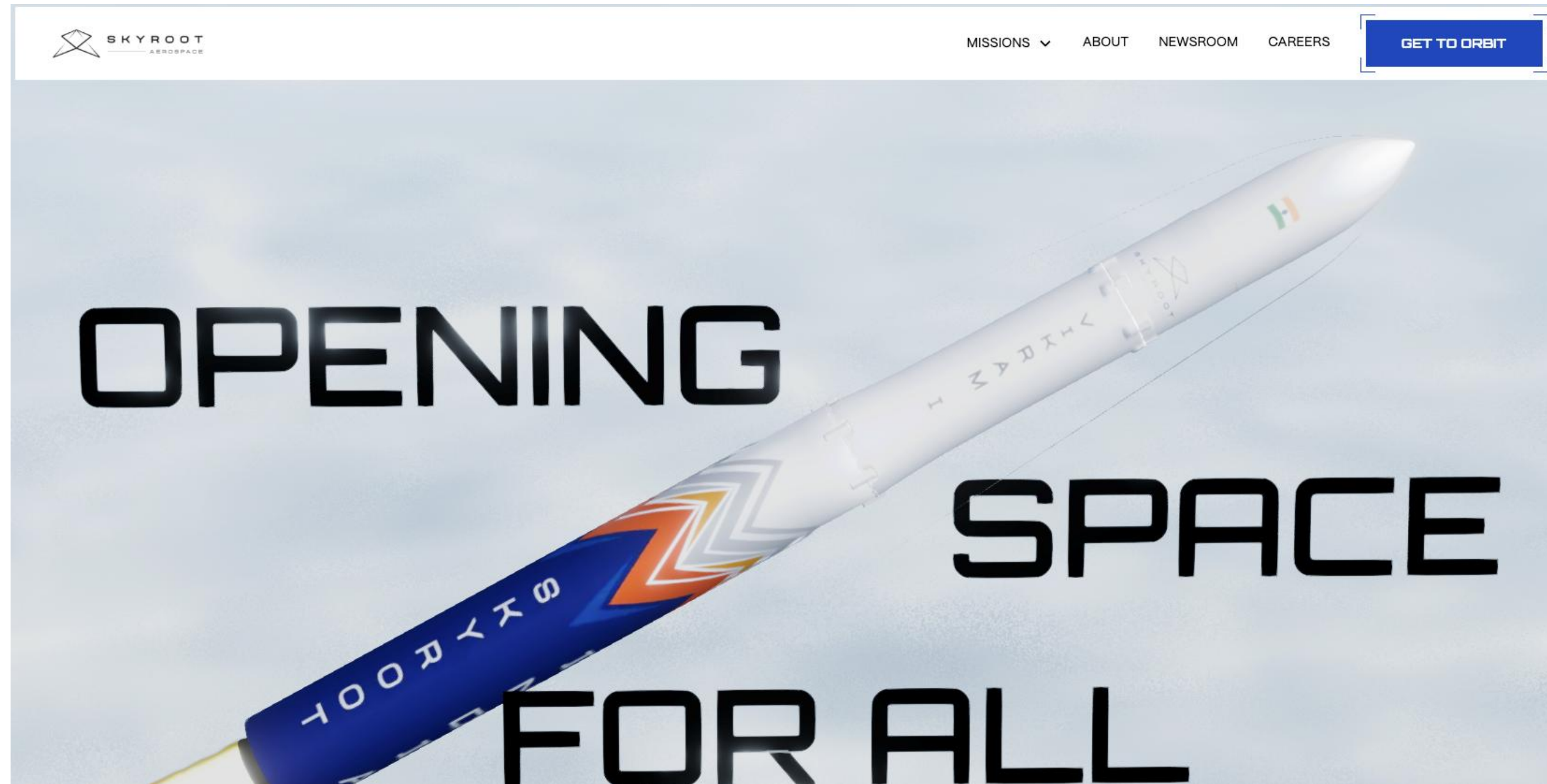




“With 31–34K start-ups, India’s tech ecosystem in 2025 is transitioning from rapid expansion to disciplined scale, marked by selective capital deployment, maturing start-up formation, and strengthening corporate-led exits”

Source: [Link](#); Nasscom-Zinnov

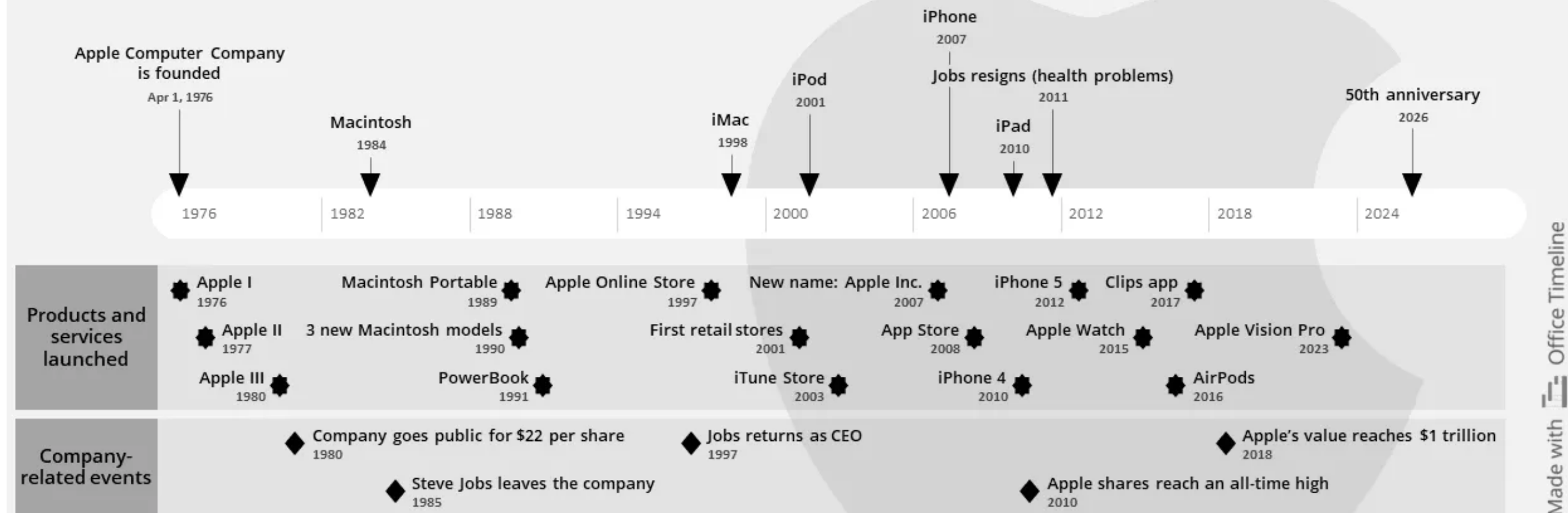
Private Space Launch (Skyroot, India)



Indian startup Skyroot Aerospace is preparing to launch Vikram-1, which could become India's first privately developed orbital rocket

Source: [Link](#); Economic Times

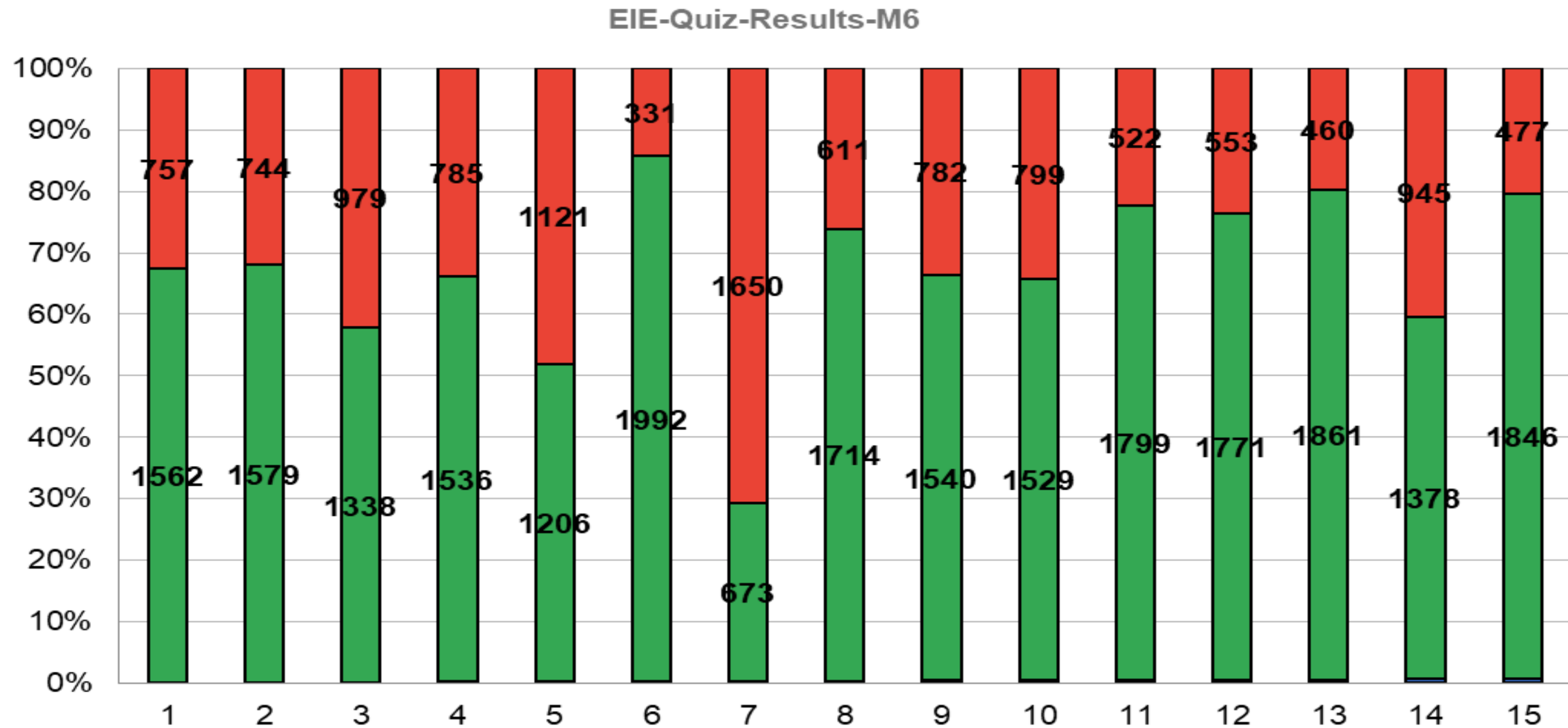
Apple history timeline



A classic example of 'startup → global company' transformation.
Shows how innovation + product design + ecosystem strategy creates enduring value.

* (April 2026)

EIE M6 Session Quiz - Discussion / Recap



EIE - II – Jan 2026 semester

EIE-II M6 Session Quiz - Discussion / Recap

Qstn #	StudentScore (Correct Answer)	StudentScore (Wrong Answer)	Total Students	% Correct	Options-wise StudentScore			
					Option A	Option B	Option C	Option D
1	1562	757	2319	67%	485	1562	159	113
2	1579	744	2323	68%	205	428	1579	111
3	1338	979	2317	58%	338	468	173	1338
4	1536	785	2321	66%	1536	225	257	303
5	1206	1121	2327	52%	209	690	1206	222
6	1992	331	2323	86%	116	118	1992	97
7	673	1650	2323	29%	551	705	673	394
8	1714	611	2325	74%	1714	207	210	194
9	1540	782	2322	66%	376	243	1540	163
10	1529	799	2328	66%	381	268	1529	150
11	1799	522	2321	78%	161	1799	231	130
12	1771	553	2324	76%	190	1771	227	136
13	1861	460	2321	80%	145	153	1861	162
14	1378	945	2323	59%	226	1378	144	575
15	1846	477	2323	79%	198	1846	143	136

EIE M6 Session Quiz - Discussion / Recap

3. In the Technology vs. Market matrix, which innovation type uses existing technology in existing markets?

- a) Radical Innovation
- b) Architectural Innovation
- c) Disruptive Innovation
- d) Incremental Innovation** - 1338 students got it right!

5. In the Problem vs. Domain Maturity matrix, when both problem and domain are NOT well defined, the innovation type is:

- a) Sustaining Innovation
- b) Breakthrough Innovation
- c) Basic Research** - 1206 students got it right!
- d) Disruptive Innovation

7. In the Technology vs. Market matrix, which innovation type introduces new technology into an existing market?

- a) Incremental Innovation
- b) Architectural Innovation
- c) Radical Innovation** - 673 students got it right!
- d) Disruptive Innovation

M6 Recap – Quiz

Menti Code: 3169 3118

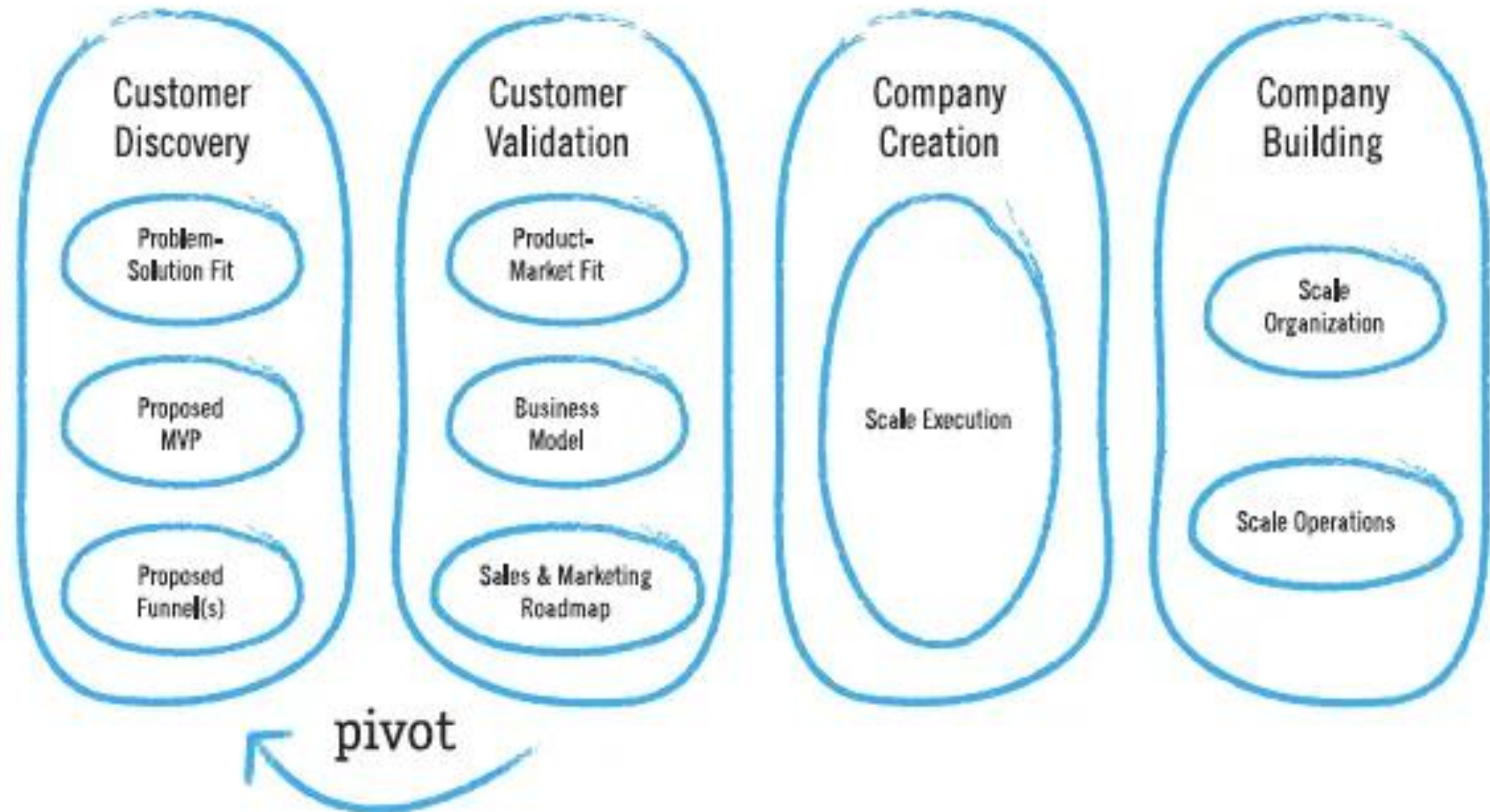


M6 Recap

- **Types of Innovation**
 - Incremental • Architectural • Radical • Disruptive
- **Technology S-Curve**
 - Rapid growth → maturity → plateau → replacement
- **Disruptive Innovation**
 - Starts niche → serves overlooked users → overtakes incumbents
- **Beyond Technology**
 - Innovation also in business models, markets, processes, experiences
- **Jobs-To-Be-Done (JTBD)**
 - Customers hire products to get a job done
- **Why JTBD Matters**
 - Better problem–solution fit → higher adoption, lower failure

Recap

Customer Development



Technology Readiness Level (TRL)



TRL	Description
9	Actual system proven in operational environment
8	System complete and qualified
7	System model or prototype demonstration in operational environment
6	Technology demonstrated in relevant environment
5	Technology validated in relevant environment
4	Technology validated in lab
3	Experimental proof of concept
2	Technology concept formulated
1	Basic principles observed

Attributed to NASA

Essentials of Innovation & Entrepreneurship (EIE) II:

Module (M7):

What is (is not) Marketing? (& why it is important for your startup)

Difference between Selling and Marketing



Selling

Getting people to exchange cash for your offering



Marketing

Satisfying the needs of customers by means of a product or service

What is **NOT** Marketing? (‘Marketing Myopia’)

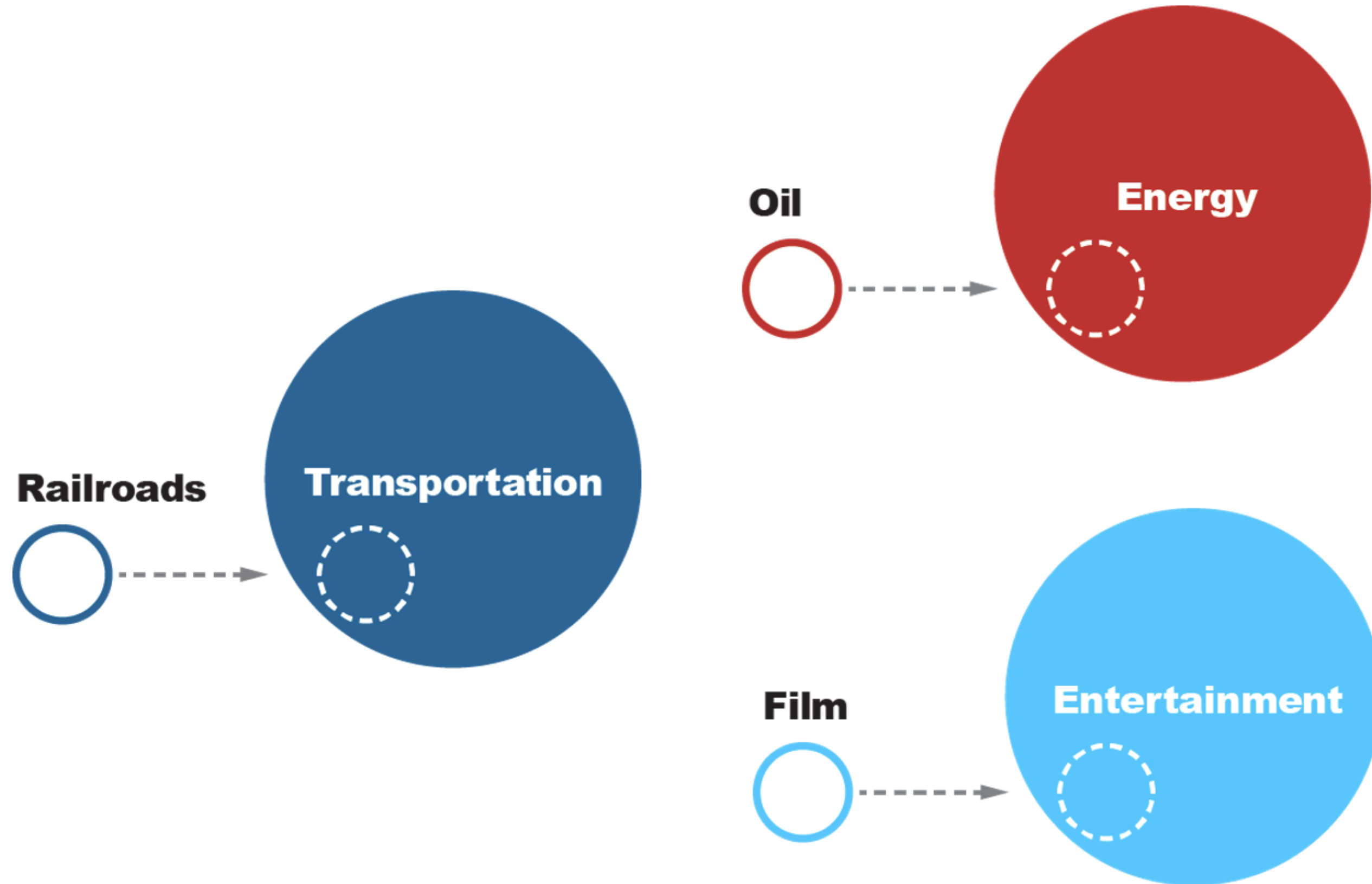


**Focusing too narrowly on
selling products and services**



**Missing the “big picture” of
What consumers really need/want**

Examples



Credit: Marketing Myopia (Harvard Business Review)

The Cure for '**Marketing Myopia**'



Think broadly about the business you are in (your product is not your business)



Create a customer-oriented company

Marketing Myopia ([link](#))

(Harvard Business Review, 2:03 m)



THE EXPLAINER: **MARKETING MYOPIA**



Harvard
Business
Review

How do you Understand the '**Industry**' you are in?

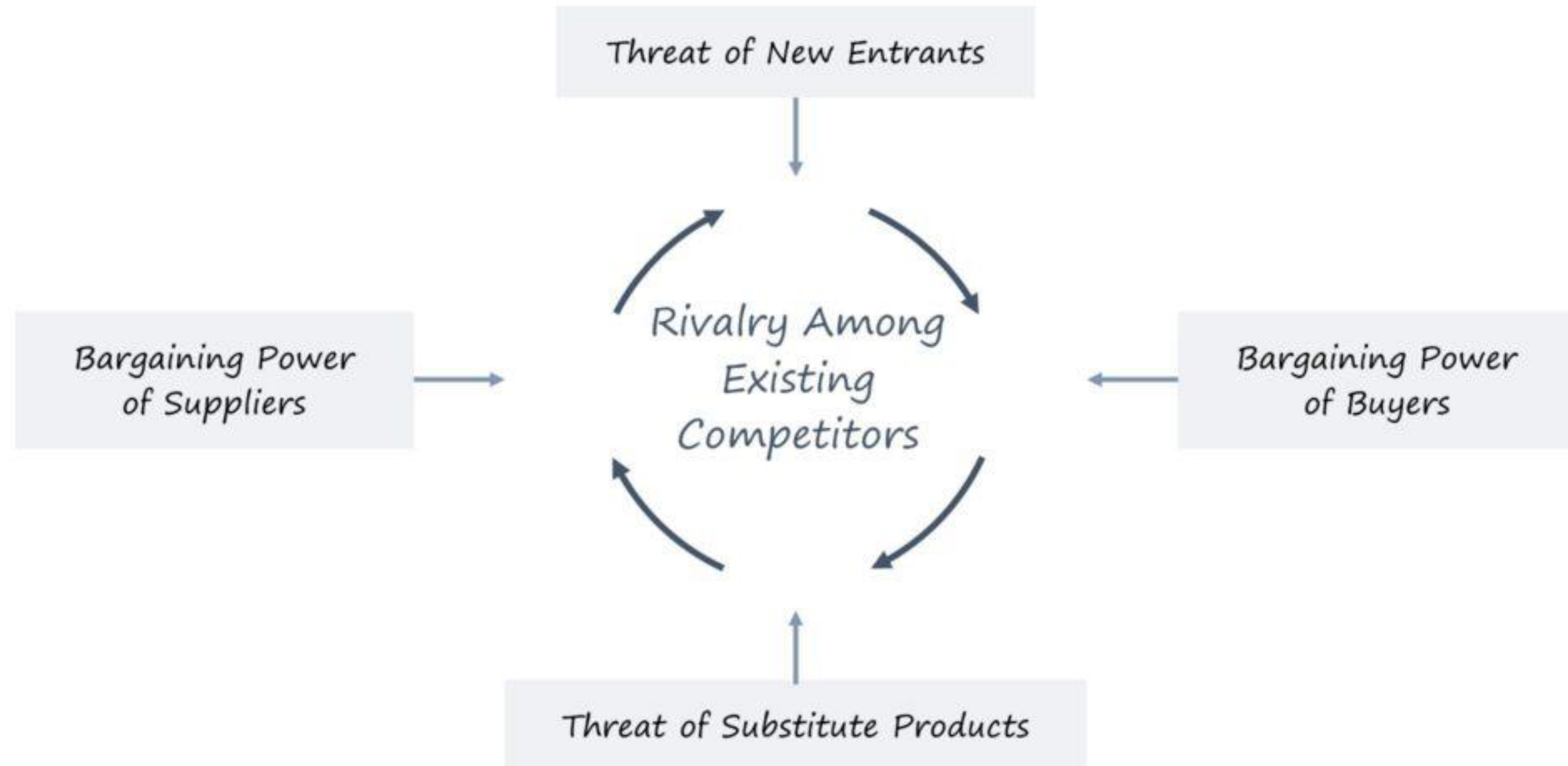


Figure Credit: WallStreet Prep (Porter's Five Force Model)

(Michael) Porter's Five Forces Model ([link](#))

(Harvard Business Review, 2:08 m)



THE EXPLAINER: **PORTER'S FIVE FORCES**



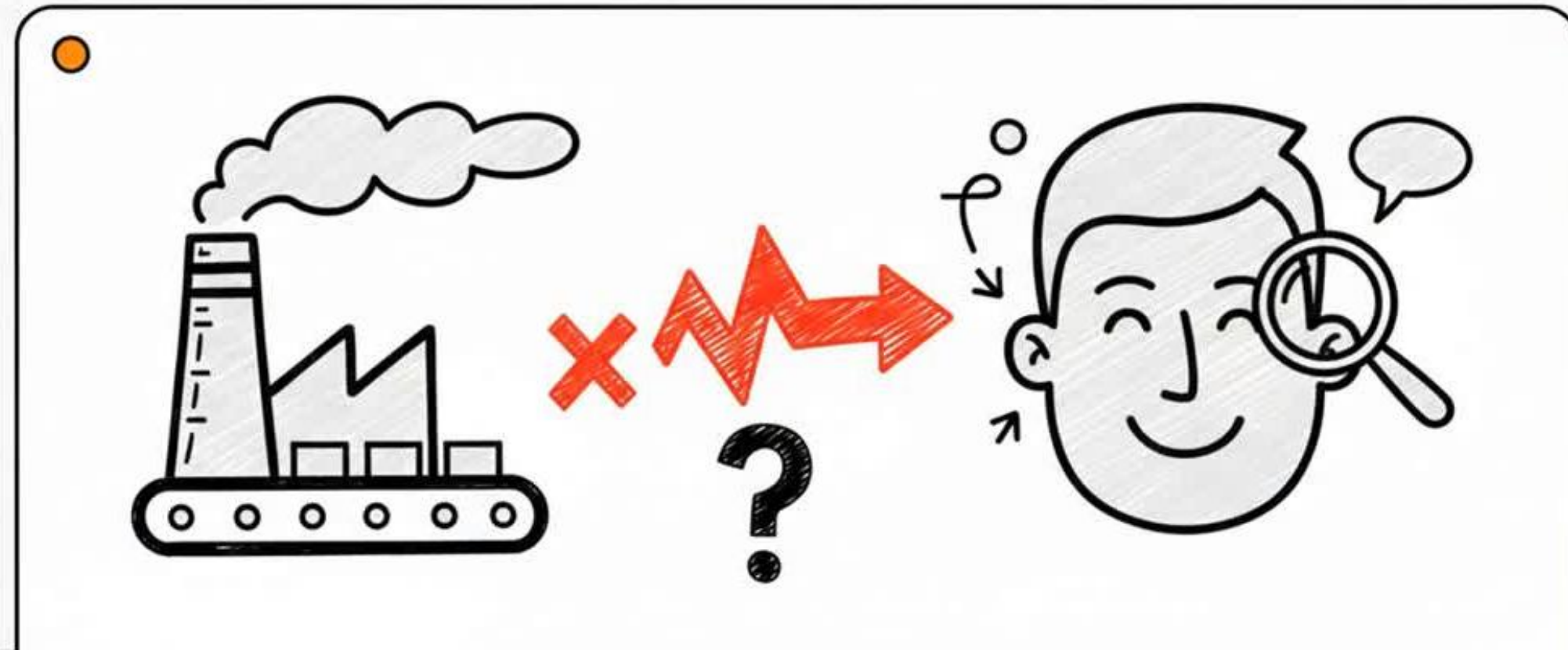
Harvard
Business
Review

Bubble Tea On Campus Venture



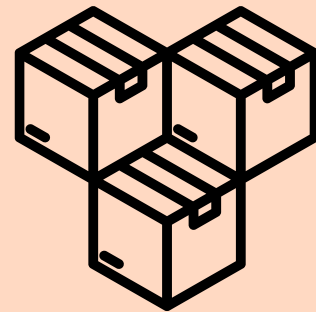
Marketing Myopia

Marketing Myopia



The Basics: 4P's of Marketing

- Features
- Quality
- Branding
- Packaging
- Gurantee



Product

- Price strategy
- Pricing
- Allowances
- Payment terms



Price

- Channels
- Locations
- Transport
- Market Coverage



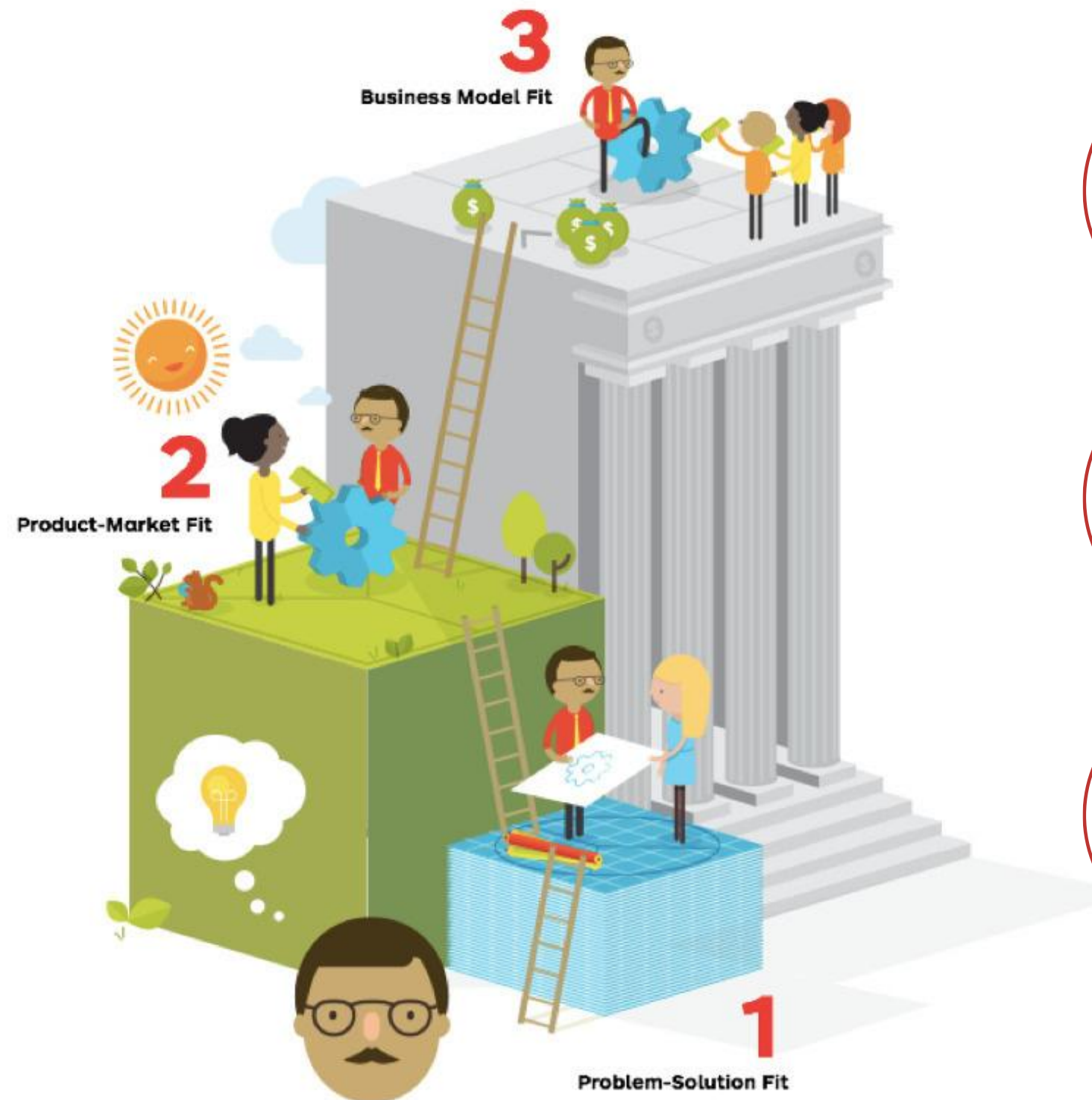
Place

- Advertising
- Promotional channels
- PR
- Discounting strategy



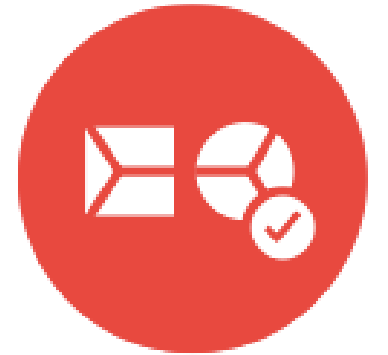
Promotion

Product Market Fit (PMF) for Startups



1

Problem-Solution Fit



2

Product-Market Fit



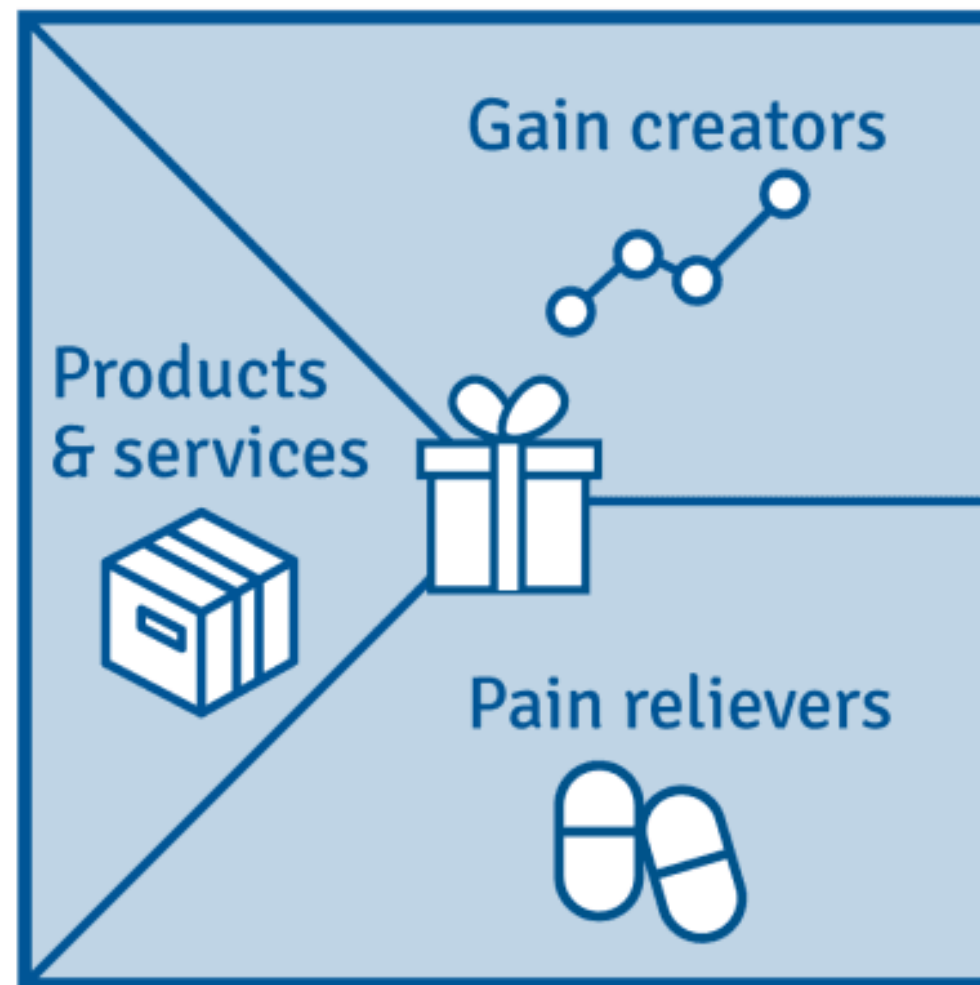
3

Business Model Fit

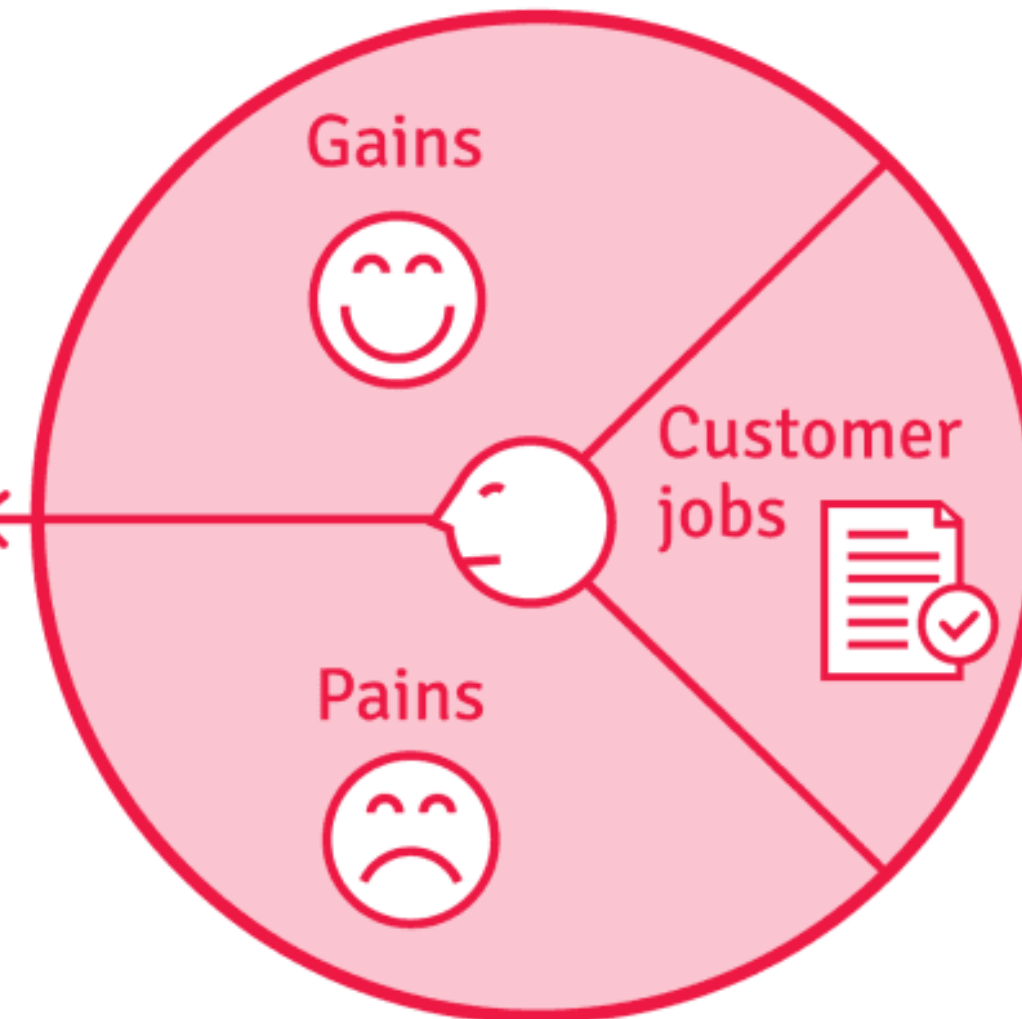


Tool: Value Proposition Canvas (Recap)

Value Proposition

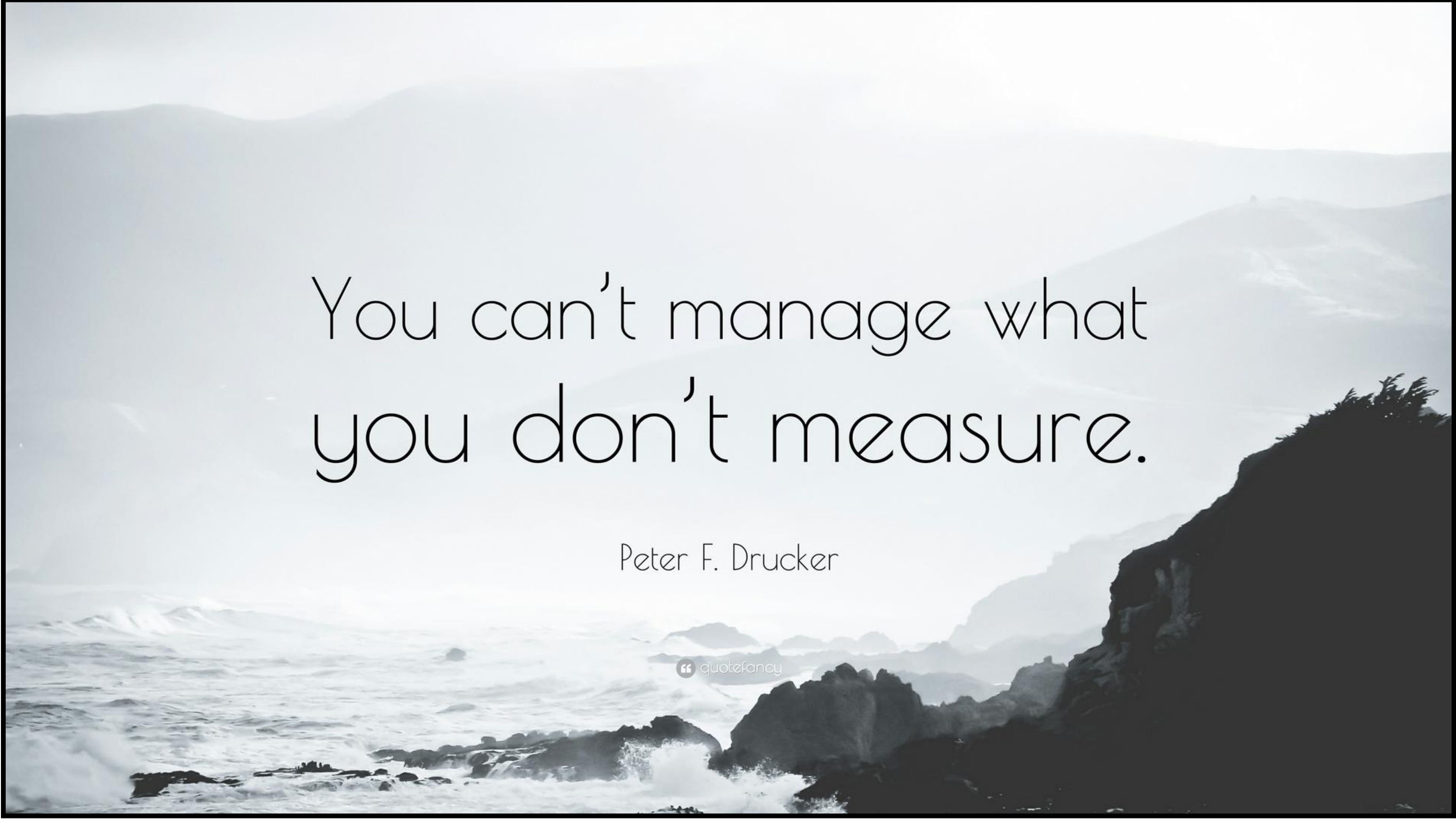


Customer Profile



Template: Positioning Statement (Recap)

For	Description of our target customer
Who	Statement of customer need or opportunity
	Name of our product
Is a	Name of the product category
That	Statement of our primary customer benefit
Unlike	Name of our primary competitor or alternative
Our	Statement of primary differentiation of product

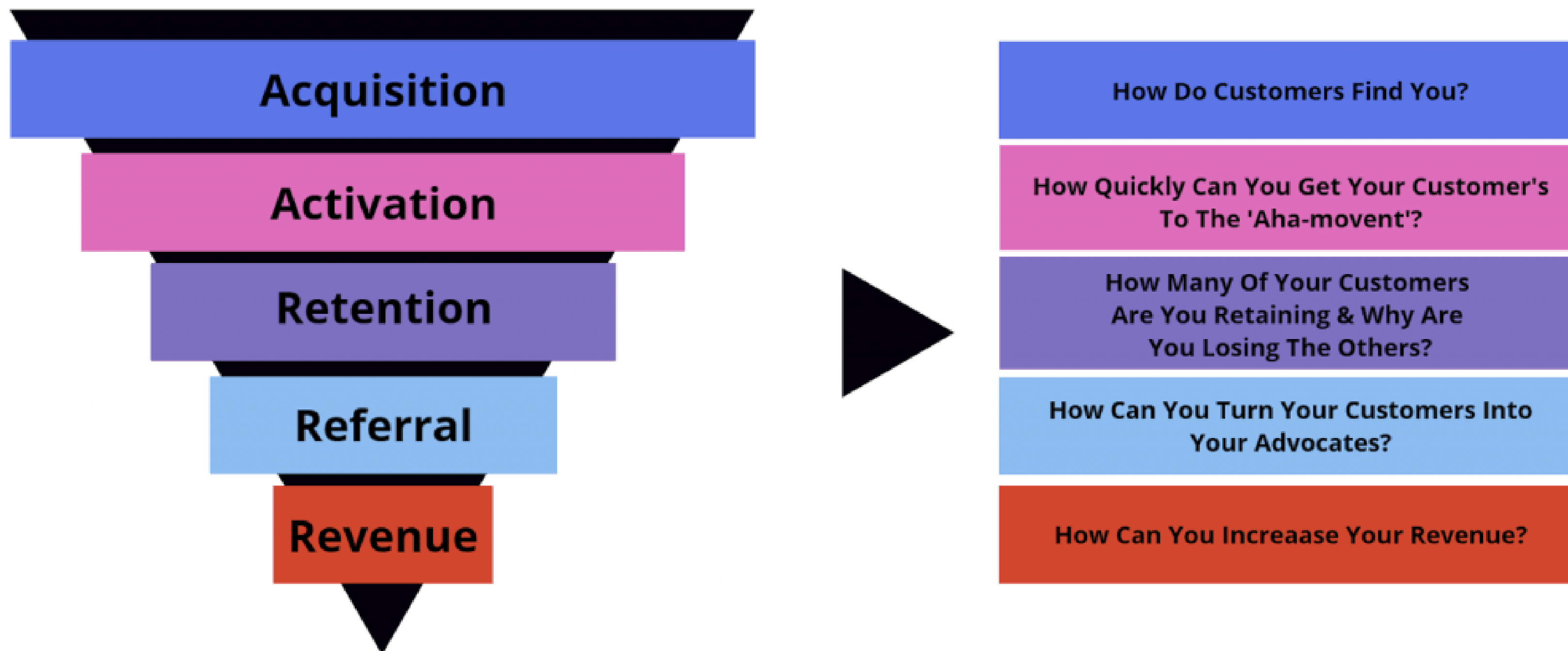


You can't manage what
you don't measure.

Peter F. Drucker

“ quote fancy

Product Market Fit (PMF) for Startups



Credit: Dave McLure, Founder 500 Startups (now 500 Global)

Startup ("Pirate") Metrics - Example

Stage	Metric	Conversion %	Estimated Value \$
Acquisition	Website visits	100%	\$0.01
Acquisition	Stays 1 min +	60%	\$0.05
Acquisition	Email captured	20%	\$1
Activation	Starts free trial	16%	\$3
Activation	Completes onboarding	25%	\$5
Retention	Visits app 5 times in 30 days	5%	\$4
Retention	Opens & clicks >2 emails	4%	\$2
Referral	Sends referral link to 1+ user	2%	\$4
Referral	1+ referred user starts trial	1%	\$10
Revenue	Generates minimum revenue	2%	\$5
Revenue	Generates breakeven revenue	1%	\$10

PMF Quiz

Menti Code: 7233 2920



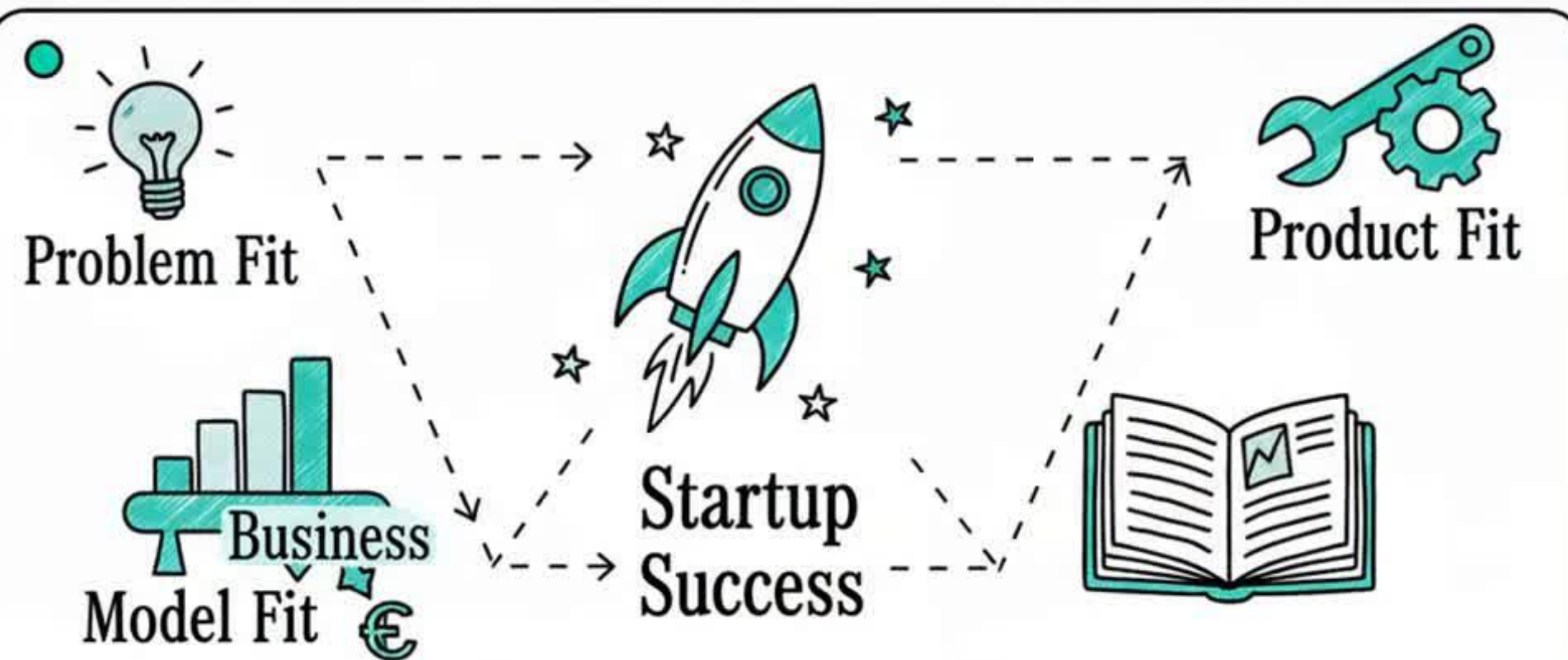
More Startup Metrics (one example)

 MRR Monthly Recurring Revenue monthly customer fees	 ARR Annual Recurring Revenue recurring revenue on an annual basis	 ARPA Average Revenue Per Account beh oisiten	 Gross Profit Total revenue minus to ot ed largest the gouds sold
 ARP Total Contract Value valley sd onlo ir an muit een	 ARR Annual Contract Value value of cnenstract over a year	 LTV Lifetime Value prediction t the net nef from toc aiomerblorers tal customer	 Deferred Revence finet ne ctor aus dution advance of in customer it
 BILLINGS orent quarter absid arderect fact charges	 CAC Customer Acquisition Cost full cost of acquiring one user	 Concentration Risk iravenure frren ontere tiic recher inp customer	 DAU Dally Active Users users other than one- tina lerpoluy
 Monthly Active Users delu rue or ne-arran on oufer & and previus chartes	 Number of logins times of sign in per month	 Activation rate number of users taking a specific action to get a a product	 MoM Growth Rate Month-or-Month Growth avaage ont a still transacting
 CMGR Compounded Abrowth nater Fllor fromti set	 Monthly Churn Rate lost customers this month/prior month total	 Activantion % of original installed base (1st month) that Decthy	 Retention (MRR intost Mom Month month/MRR at the worth transacting
 Netn (MRR lost MRR from upsells) brur ale s t U2 Hest 1/28 month	 Burn Rate Monthly cash burn rate	 TAM Total Addressable Market revenue opportunity pr elualuct	 MRR Projection projection of current MRR into the at still annualized

Credit: agrassoblog

Product Market Fit (PMF)

Finding Product-Market Fit



Essentials of Innovation & Entrepreneurship (EIE) - II

Module (M7): (AI-driven) Market Research

Why do **Market Research**?

When? When not to?

Steve Jobs on Market Research ([link](#))

(YouTube, 1:26 m)



Market Research: The Secret Ingredient for Business Success ([link](#))

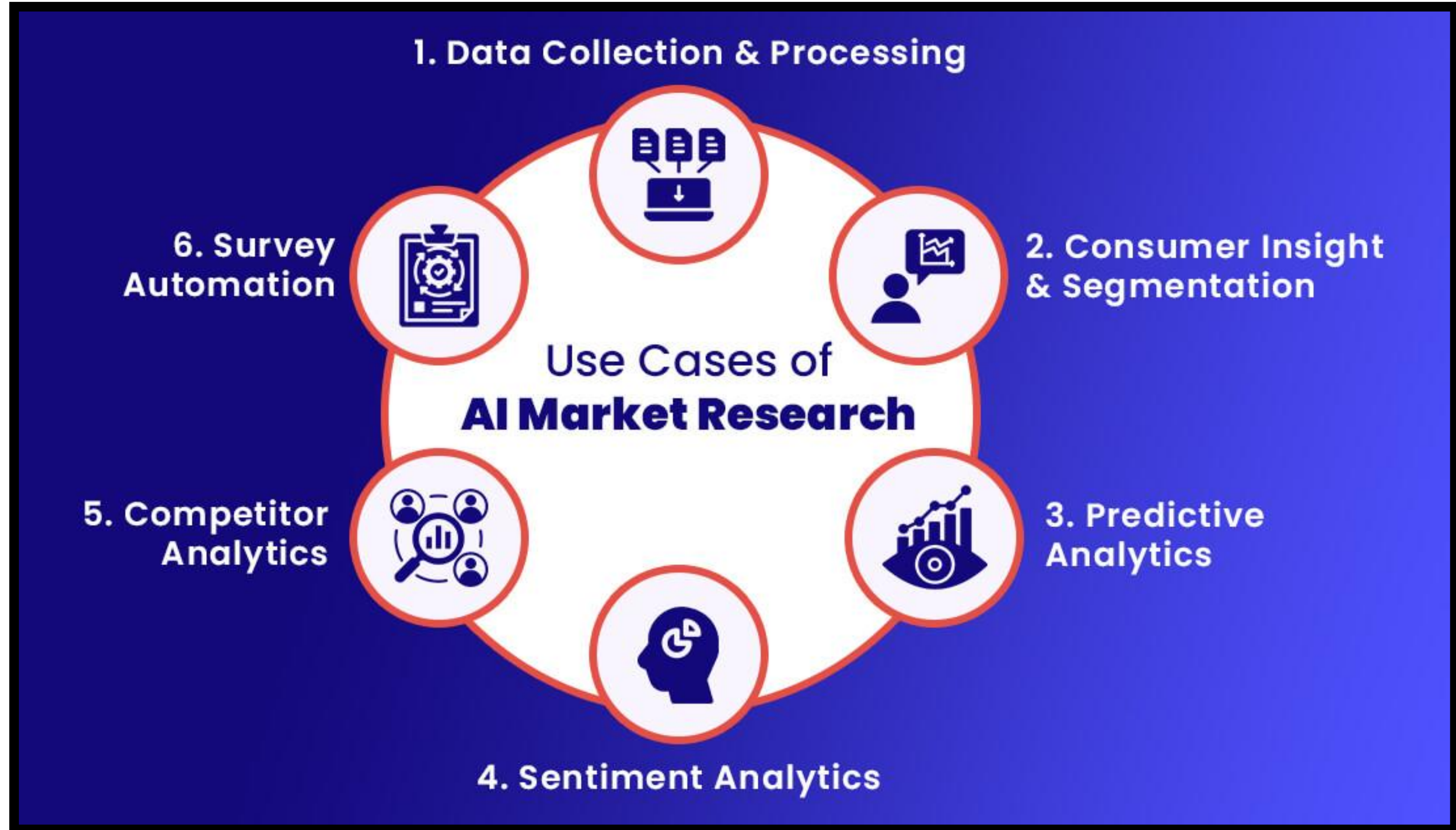
(Leaders Talk, 5:13 m)



Why AI based Market Research?

Aspect	Traditional Market Research	AI-Augmented Market Research
Speed & Efficiency	Manual processes that are time-consuming.	Rapid, automated analysis delivering real-time insights.
Cost	Expensive due to extensive human labor.	More cost-effective through automation and scalability.
Data Collection	Limited sample sizes and slower data gathering.	Broad, diverse, and continuous data collection from multiple sources.
Accuracy & Bias	Prone to human bias and error.	Objective analysis with reduced bias and improved accuracy.
Adaptability	Provides retrospective insights with delayed updates.	Offers predictive analytics for proactive decision-making.

AI Augmented Market Research



AI in Market Research

Sentiment Analysis	Predictive Analysis	Social Media Listening	Chatbots and Virtual Assistants
Customer Journey Analysis	Demand Forecasting	Customer Segmentation	Image, Video and other media analysis

AI Tools for Market Research: Examples

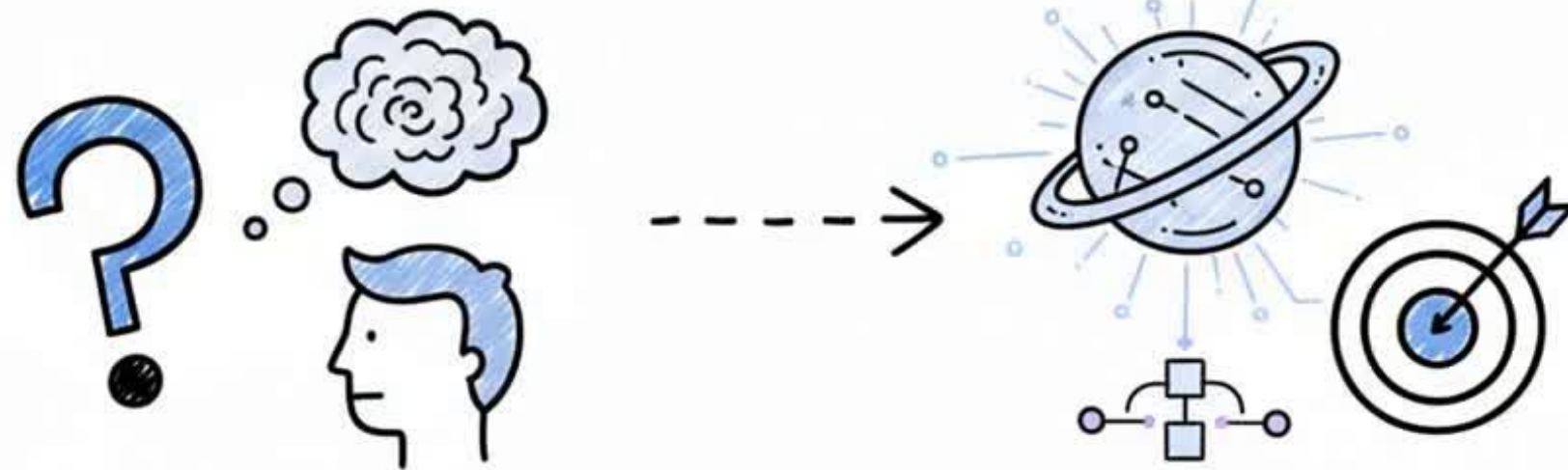
Activity	Tool
General Market Research	Perplexity, ChatGPT, Google Gemini, Claude and many more Custom GPTs
Competitor Analysis	Crayon.ai, Brand24, Kompyte AI etc
Quantitative Research	SEMRush Market Explorer, Survey Sparrow, Algolia
Qualitative Research	Insight7, Atlas.ti

Key Takeaways

- **AI + HI + Market Research = Better Outcomes**
- **AI = Better, Cleaner, Up-to-date Data = Better Decisions**
- **AI cannot replace Human Cultural nuances, Emotions**
- **AI is as good as the Data it is trained on**
- **Initially can be complex and costly but will give results in time**
- **AI Algorithm Bias**
- **Context Windows and Misinterpretation**

AI in Market Research

Guesswork to Prediction





For Tomorrow's Innovators

CIE Ignite Season 02: A curriculum-integrated innovation program delivering to 3,200+ engineering and non-engineering students at your most motivated moment.

From classroom theory to real-world impact!

Phase 2 Status

105

Teams

64

Responded

11

Off Radar

3

PPT Ready

Phase 2 Status at a Glance

64 / 105

Teams Responded
61% Response Rate

56 / 64

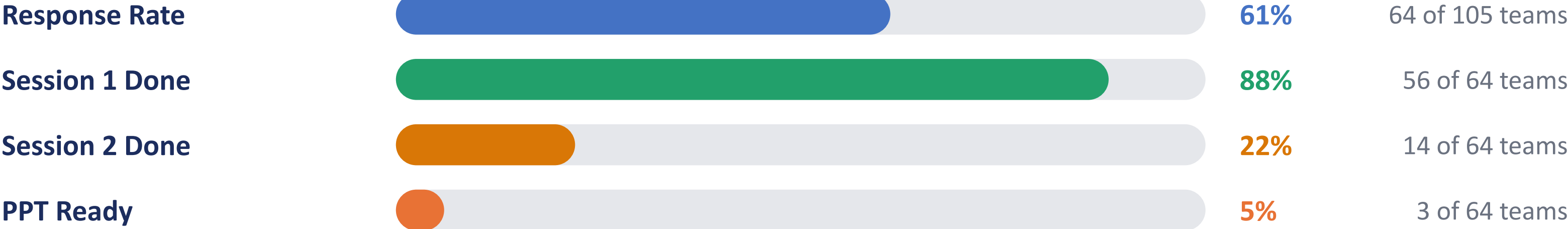
Completed Session 1
88% of respondents

14 / 64

Completed Session 2
22% — needs attention

3 / 64

Pitch Decks Ready 🚀
5% — but the best 5%!



What You're Telling Us

What's Working Well

Mentoring very helpful

~42 teams

Great learning experience

~28 teams

Want more mentor sessions

~8 teams

Structured phases work well

~6 teams

What to Improve

ISA timing clash

~15 teams

More time between phases

~11 teams

Scheduling & comms gaps

~7 teams

Better mentor-domain match

~3 teams

" The mentoring session helped us refine our project idea and think from a real startup perspective — not just a technical implementation. "

New Week: Pitch Day!

March 16, 17 & 18 · CIE, MRD 1st Floor · Your moment to shine



Complete Session 2 TODAY

Contact your mentor right now — don't go into pitch day without mentorship done.



Finalise Your Pitch Deck by March 12th

Submit by 12 Noon on March 13th . Make sure slides are crisp, data is validated, story is clear.



Off-Radar Teams: Respond Immediately

11 teams have zero contact. Reach out to cie@pes.edu right now — or risk elimination.



Come Ready to Pitch with Confidence

Faculty, mentors & partners will be judging. This is your moment. Make it count!

Innovation starts here—and it doesn't stop.

Our Website: cie.pes.edu

Reach out: tarunrama@pes.edu

Faculty Talk : Dr. V. Krishna

Dean, Student Affairs, PES University,
Mridangam Vidwan & accomplished
musician





Background

- › 28+ years in teaching, research, consultancy & administration
- › Ph.D. — Visveswaraya Tech. Univ., Belgaum (Cryogenic Heat Exchangers, 2013)
- › M.E. — BIT Mesra (1993) | B.E. — Bangalore University (1989)



Responsibilities at PES

- › Domain Lead, Thermal Engineering Domain
- › Chief Course Instructor: Thermodynamics, Mechanics of Fluids, Heat Exchanger Modelling
- › Dean, Student Affairs — festivals (Aatmatrisha, Maaya, PECON, PES Debate), welfare & emergencies



International & Academic Engagement

- › Visiting Professor, Tecnológico de Monterrey, Mexico (Aug 2018)
- › 24+ publications (10 journals, 14 conferences) | Invited speaker & chair at intl. conferences
- › Reviewer: Heat Transfer Engineering (Taylor & Francis), Int'l J. of Heat & Mass Transfer (Elsevier)
- › Patent: Advanced Coolants with Nanoparticles (Indian Patent #201741044247, 2017)



Beyond Academia

- › Accomplished Mridangam artist; performs with leading Carnatic musicians nationally
- › Executive Director, Percussive Arts Centre
- › Faculty Head — Team Haya (Racing Team), NINAADA (Indian Culture Forum)

Founder Talk : Murli Thirumale

“Serial Entrepreneur”
**(His First Business Was Acquired By Citrix,
His Second By Dell, And his Third by Pure
Storage after having raised \$50 Million!!)**



Founder Talk • Murli Thirumale

B.Tech IIT BHU • MBA Kellogg School of Management



HP – General Manager



Net6 – Co-founder & CEO



Citrix – SVP & GM



Ocarina Networks - Co-founder and CEO



Dell – Executive



Portworx – Co-founder & CEO (acquired by Pure Storage)



Pure Storage (now Everpure) – GM

3x

Co-founded
companies

25+

Years in
tech & marketing

3

Acquisitions

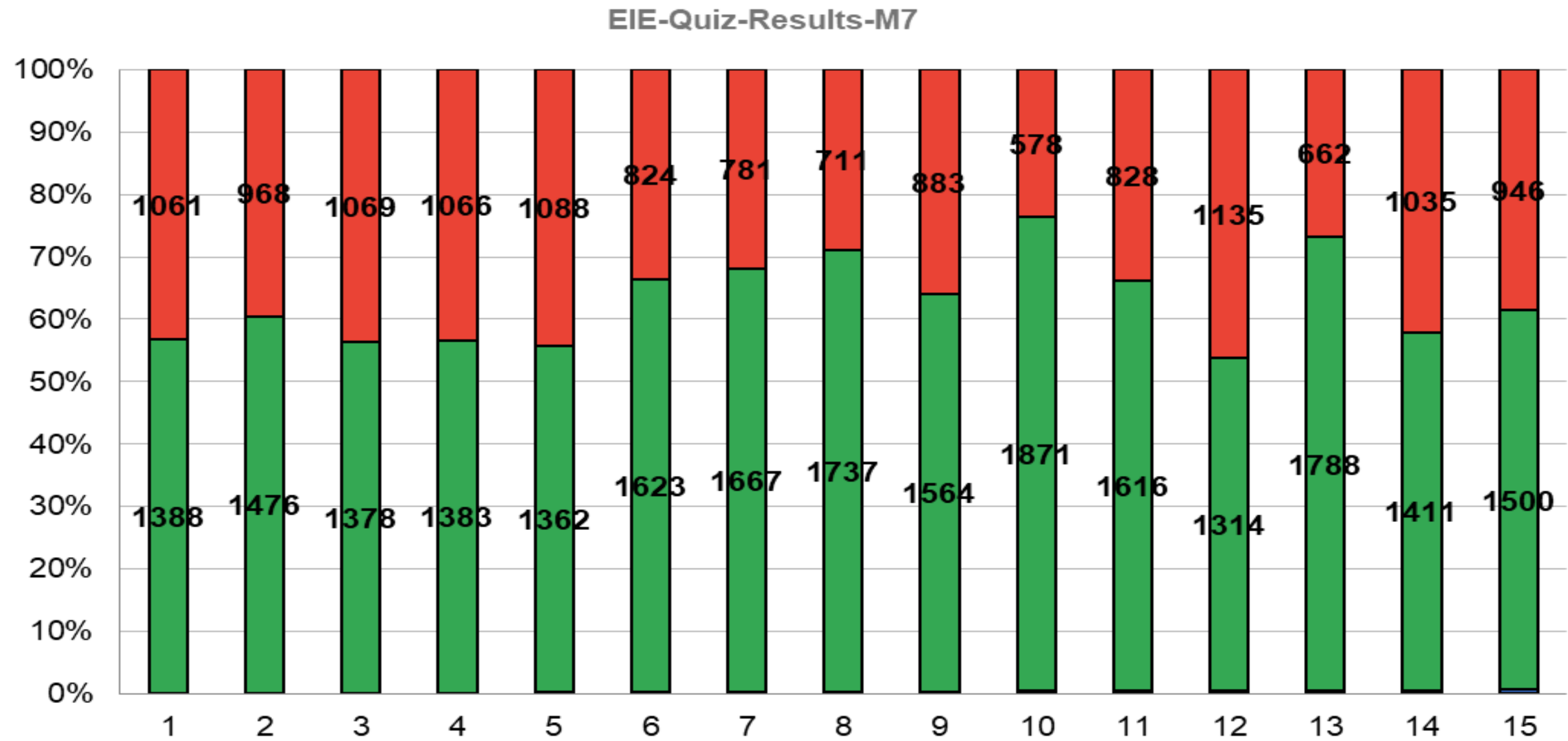
Summary

- Marketing is NOT Selling
- Marketing is understanding customer needs and designing solutions to meet them
- Marketing goes beyond your product and competitors — it requires understanding the industry ecosystem
- For Startups
- A defining milestone is Product–Market Fit
- Understand why, what, and how customers adopt your product
- Drive critical decisions using startup metrics, not opinions
- Adopt AI where it fits for market research and marketing

Essentials of Innovation & Entrepreneurship (EIE) - II

Module (M8): Startup Pitch – Why/What/How Prototyping for Startups

EIE M7 Session Quiz - Discussion / Recap



EIE-II M7 Session Quiz - Discussion / Recap

Qstn #	StudentScore (Correct Answer)	StudentScore (Wrong Answer)	Total Students	% Correct	Options-wise StudentScore			
					Option A	Option B	Option C	Option D
1	1388	1061	2449	57%	567	1388	193	301
2	1476	968	2444	60%	435	360	1476	173
3	1378	1069	2447	56%	179	1378	677	213
4	1383	1066	2449	56%	651	262	1383	153
5	1362	1088	2450	56%	1362	392	447	249
6	1623	824	2447	66%	197	1623	402	225
7	1667	781	2448	68%	282	1667	164	335
8	1737	711	2448	71%	193	1737	276	242
9	1564	883	2447	64%	577	137	1564	169
10	1871	578	2449	76%	123	1871	226	229
11	1616	828	2444	66%	1616	528	152	148
12	1314	1135	2449	54%	375	232	1314	528
13	1788	662	2450	73%	1788	262	196	204
14	1411	1035	2446	58%	637	143	255	1411
15	1500	946	2446	61%	593	1500	172	181

EIE M7 Session Quiz - Discussion / Recap

3. A company that defines itself only by its product and not by the broader customer need is showing:

- a) Market expansion
- b) Marketing myopia** - 1378 students got it right!
- c) Product differentiation
- d) Customer discovery

5. A new ride-sharing startup worries that customers may switch to metro trains instead of using its service. Which force is being analysed based on Porter's Five Forces?

- a) Threat of substitutes** - 1362 students got it right!
- b) Rivalry among competitors
- c) Bargaining power of buyers
- d) Threat of new entrants

12. A startup defines itself as being in the "oil business" instead of the "energy solutions business." This is an example of:

- a) Customer segmentation
- b) Market penetration
- c) Marketing myopia** - 1314 students got it right!
- d) Value proposition design

M7 Recap: Marketing - Quiz

Menti Code: 51019750



M7 Recap

1) Marketing is not selling.

- Marketing focuses on understanding customer needs, not just pushing products for sales.

2) Avoid marketing myopia.

- Define your business by the problem you solve, not the product you sell.

3) Understand the industry ecosystem.

- Tools like Porter's Five Forces help analyze competition and market dynamics.

4) Product Market Fit is critical for startups.

- Startups must achieve problem solution fit before scaling the business model.

5) Metrics drive better decisions.

- Metrics like AARRR and others help track growth, retention, and revenue effectively.

6) AI improves market research.

- AI enables faster, data-driven insights but must be combined with human judgment.

What is Persuasion?

persuasion noun

pər-'swā-zhən 

Definition of *persuasion* >

1 as in *convincing*

the act of reasoning or pleading with someone to accept a belief or course of action
the suffragists' gradual *persuasion* of the American people that voting rights had to be extended to women

What is Persuasion?

Synonyms & Similar Words

Relevance 

convincing

inducement

inducing

cajolery

entreaty

pressuring

urging

cajolement

conversion

suasion

lobbying

blandishment

influencing

prompting

tempting

overpersuasion

persuading

seduction

coaxing

wheedling

exhortation

swaying

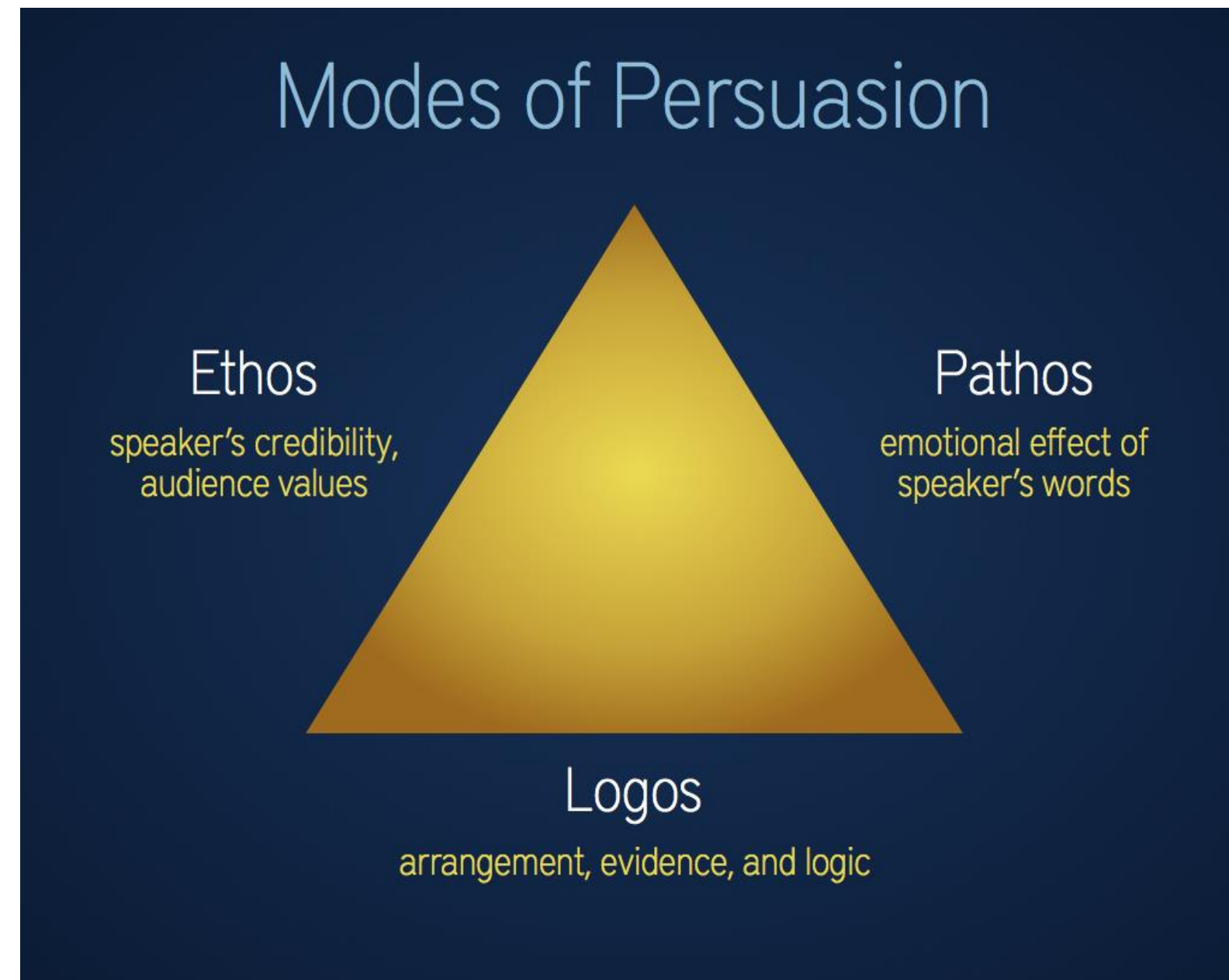
brainwashing

Credit: Merriam Webster Dictionary

The Pitch – It is all About Persuasion

Ethos: Build Credibility

- **Expertise:** Your experience & skills.
- **Trust:** Honesty & authenticity.
- **Authority:** Reputation & sources.
- **Values:** Shared goals & principles.



Pathos: Create Connection

- **Storytelling:** A relatable narrative.
- **Emotion:** Language that inspires feeling.
- **Needs:** Address audience pain/hopes.
- **Passion:** Your genuine enthusiasm.

Logos: Prove Your Point

- **Data:** Facts, stats, & numbers.
- **Structure:** A clear, rational argument.
- **Evidence:** Demos, case studies, proof.
- **Clarity:** Simple, understandable reasoning.

Elements of a Startup Pitch

- **Motivation:** Why are you here? What is your 'WoW' statement?
- **Problem Statement:** What are you solving?
- **Solution:** How are you solving the problem? (aided by Prototype, MVP etc)
- **Traction:** How is the startup doing? Share insights about product usage, customers, etc
- **Market:** Representation of your market, position in the market, competition, etc
- **Revenue Model/BMC:** Financials and key related elements
- **Teams:** Who are the team members and why is this a 'dream team'?
- **Call-to-Action (CTA):** What actions do you want the audience to perform?

Pitching: Key Aspects

1. Focus on your Audience
2. Story Telling / Narrative
3. Use Data Effectively
4. Visual Aspects
5. Deliver with Presence

Focus on Your Audience

Who are **They**?

Why are **They** here?

What is in it for **Them**?

Focus on Your Audience

- Opening Your Presentation (“Hook”) ~30 seconds
- Grab the audience’s attention
- An intriguing question, Surprising Fact, Compelling Story
- State the Purpose and Objectives

**“Think of your presentation as a rocket 🚀 on a launchpad
Not an airplane ✈️ on a runway”**

Data Visualization

- Graphical representation of information and data
- Use visual elements (charts, graphs, maps, etc)
- Show/discuss trends, outliers, and patterns in data
- Simplify complex data without losing the true meaning

The Perfect Pitch Blueprint



Data Visualization (Examples)



Note: The original is interactive

Credit: US Office of Management and Budget (2016)

Data Visualization (Examples)

NEXT AMERICA

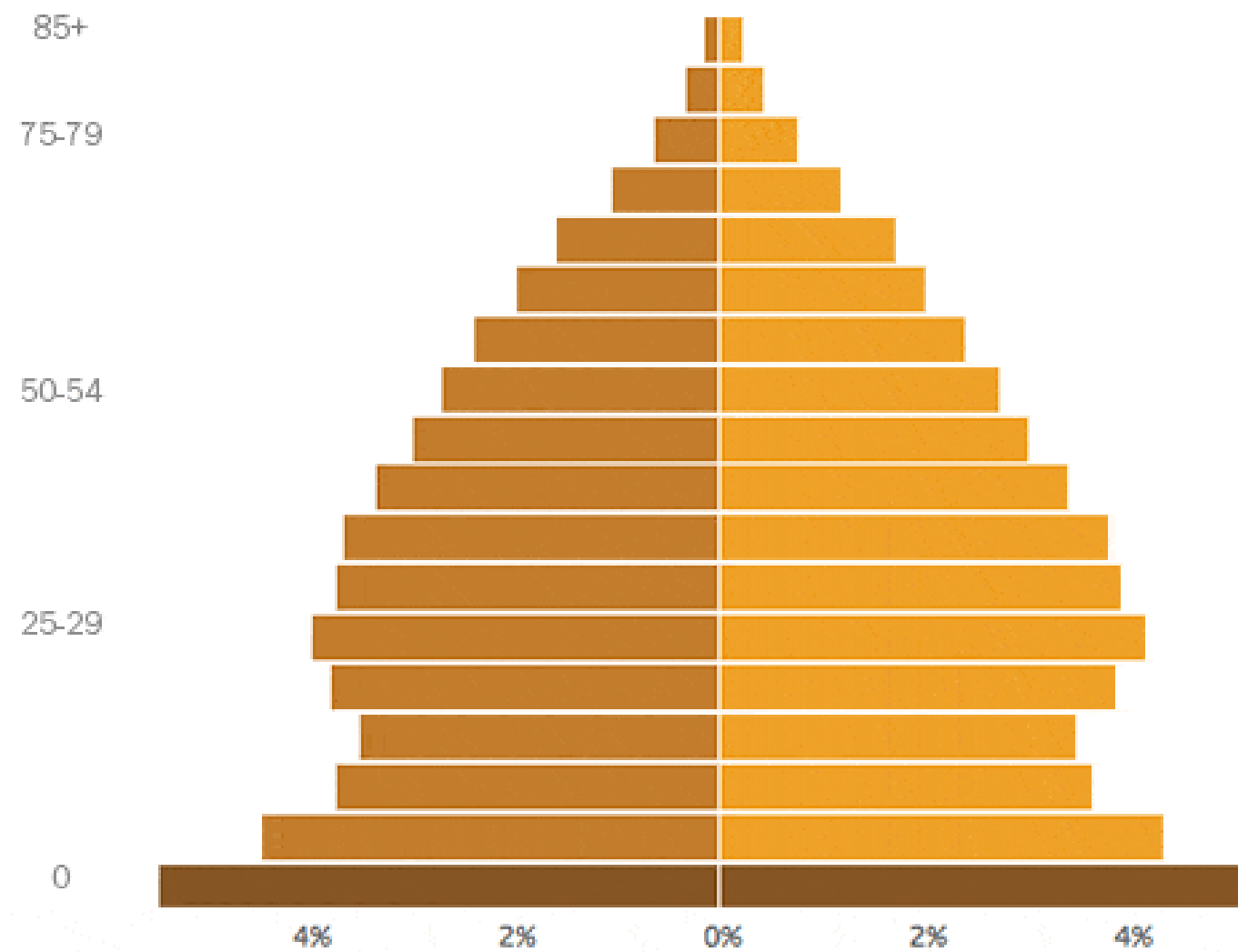
Percent of U.S. Population by Age Group, 1950-2060

■ Baby Boomers

MALE

1950

FEMALE



PEW RESEARCH CENTER

Credit: Pew Research Center

Data Visualization (Examples)

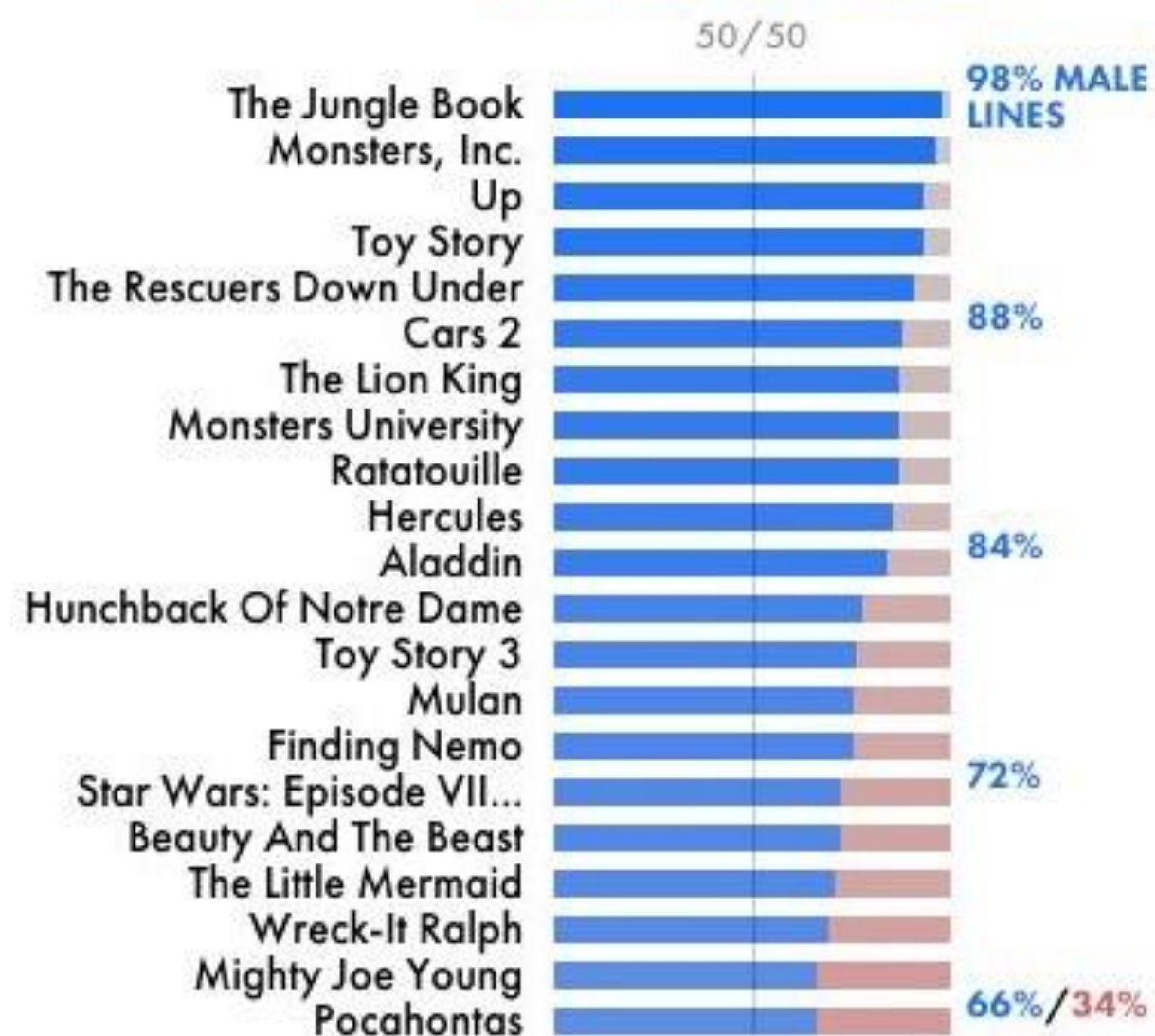


Screenplay Dialogue, Broken-down by Gender

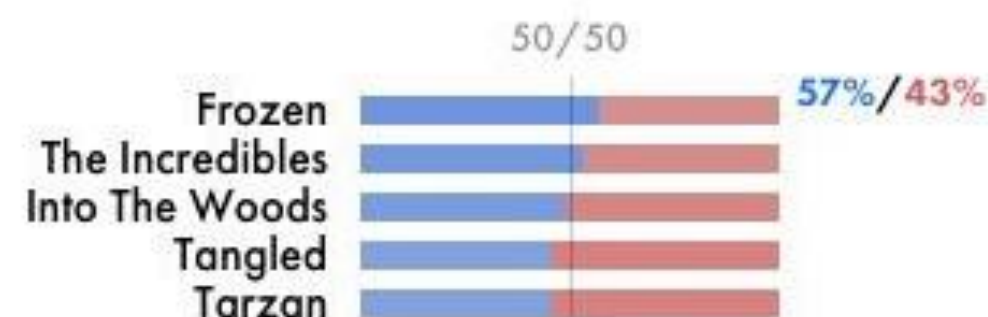
2,005 Screenplays: Dialogue
Broken-down by Gender

Only High-Grossing Films: Ranked in
the Top 2,500 by US Box Office*

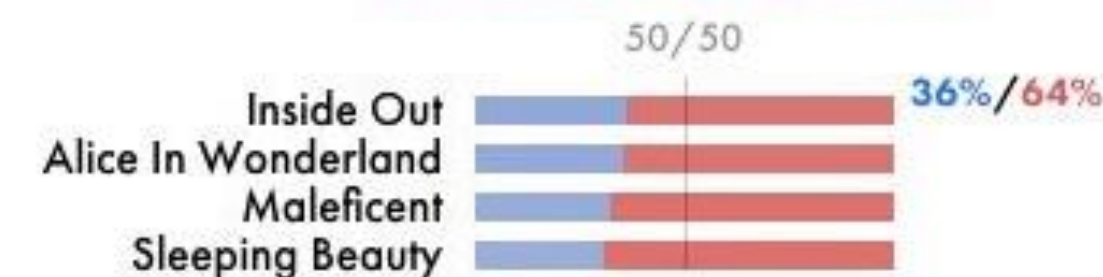
Men have 60%+ Lines



Gender Balance, +/- 10%



Women have 60%+ Lines



Credit: Pew Research Center

Visual Aspects: Fonts

- Readability and Aesthetic
- Keep it Simple: Sans Serif Fonts (Arial, Helvetica)
- Consistent font usage (1-2 font styles: Heading, Body)
- Font size (big, easy to read: 30 pt; bigger for Heading)

Visual Aspects: Color



New Customer Acquisition in India

Total Acquisitions:
123,456,789

YTD Goal:
150,000,000





New Customer Acquisition in India

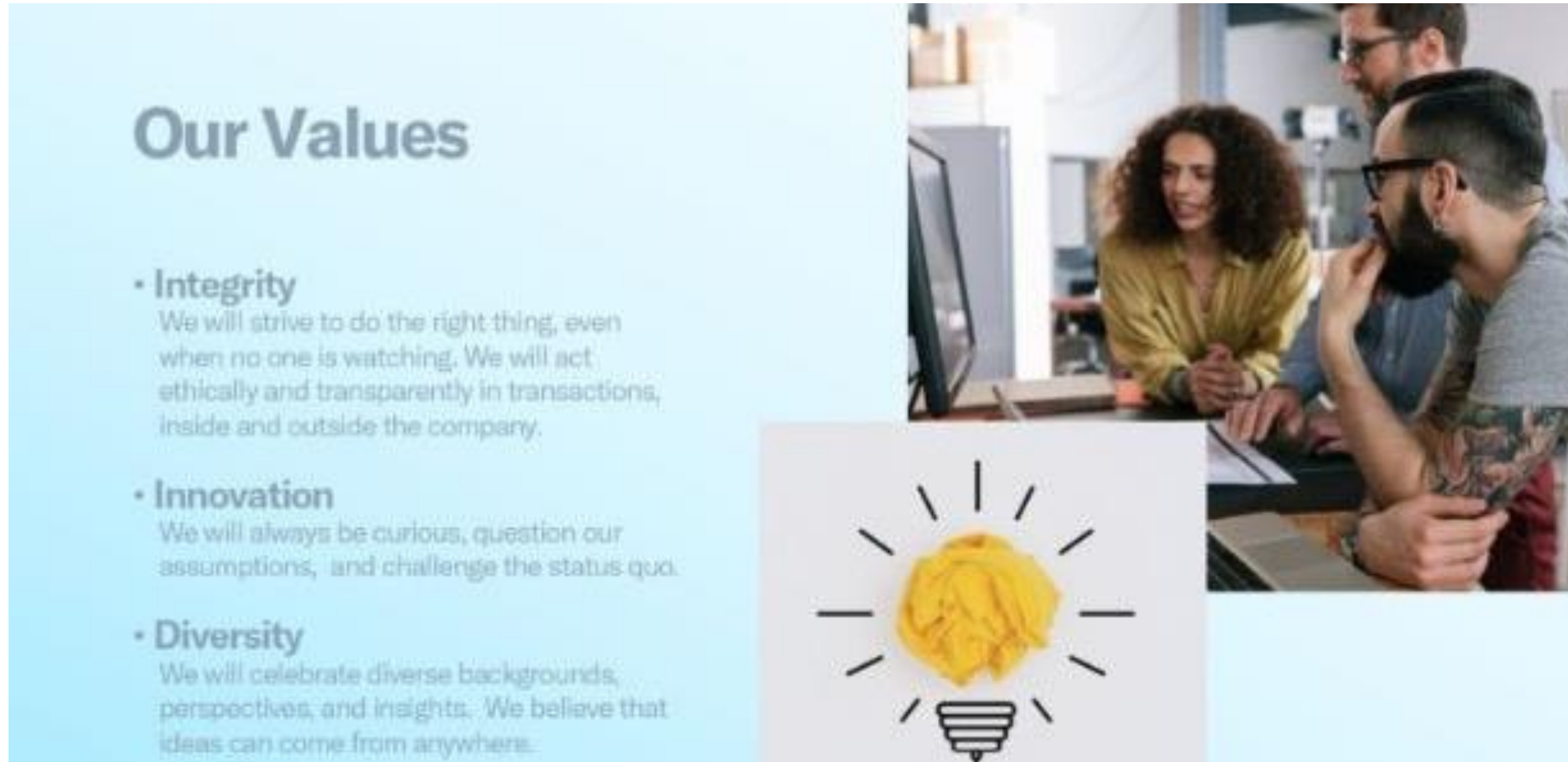
Total Acquisitons:
123,456,789

YTD Goal:
150,000,000



A limited palette of 2-3 main color, Simple background

Visual Aspects: Composition



Visual Aspects: Composition



Our Values

- Integrity
- Innovation
- Diversity



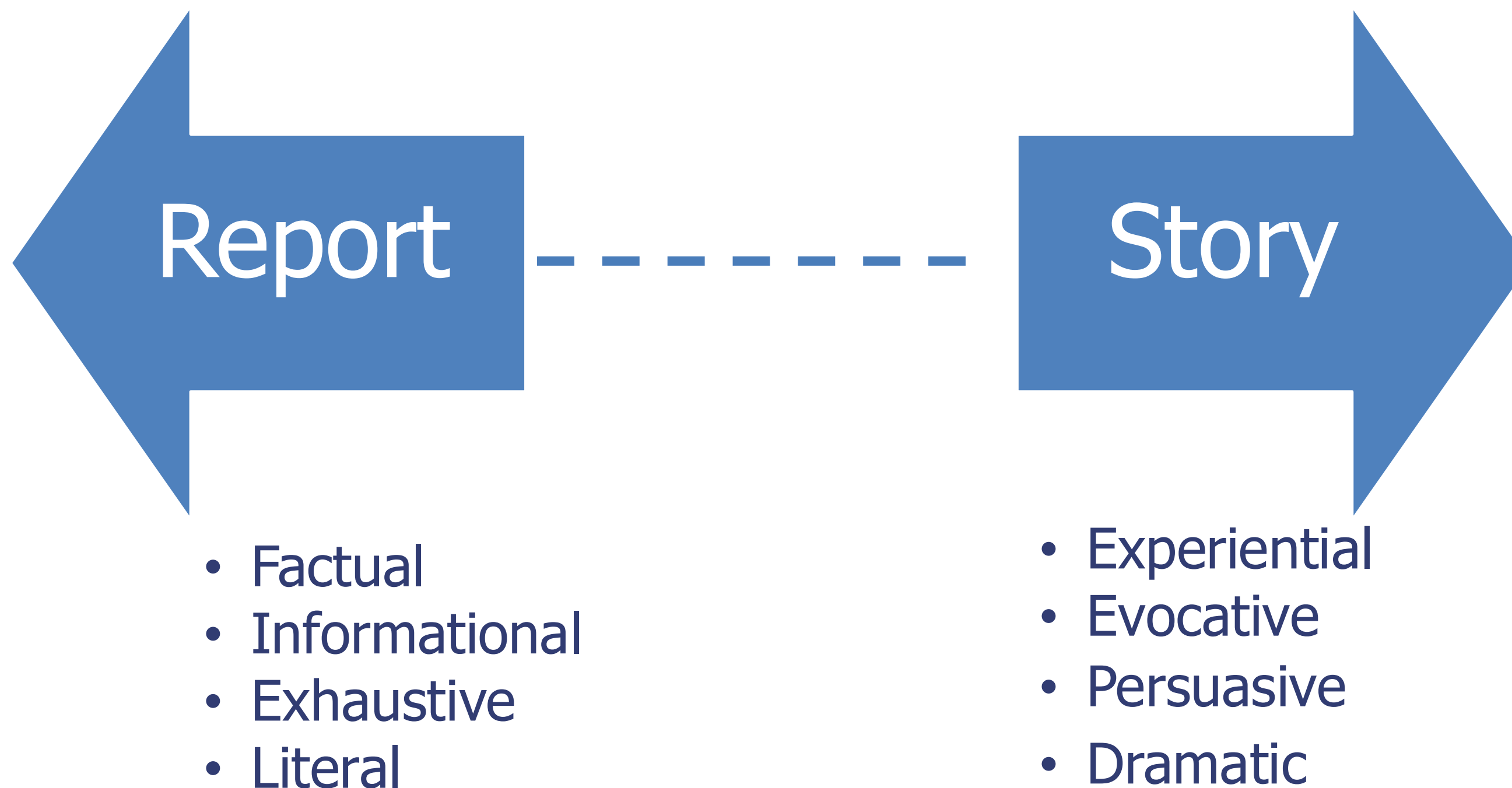
Credit: How to Make a “Good” Presentation “Great” (Harvard Business Review)

Story Telling / Narrative

- How will you frame your story?
- What is the best way to connect with the audience?
- Storytelling is not a speech; It involves active listening.
- Good pitches combine both data & effective story telling.

Storytelling

Find the Right Mix of *Data* and *Narrative*



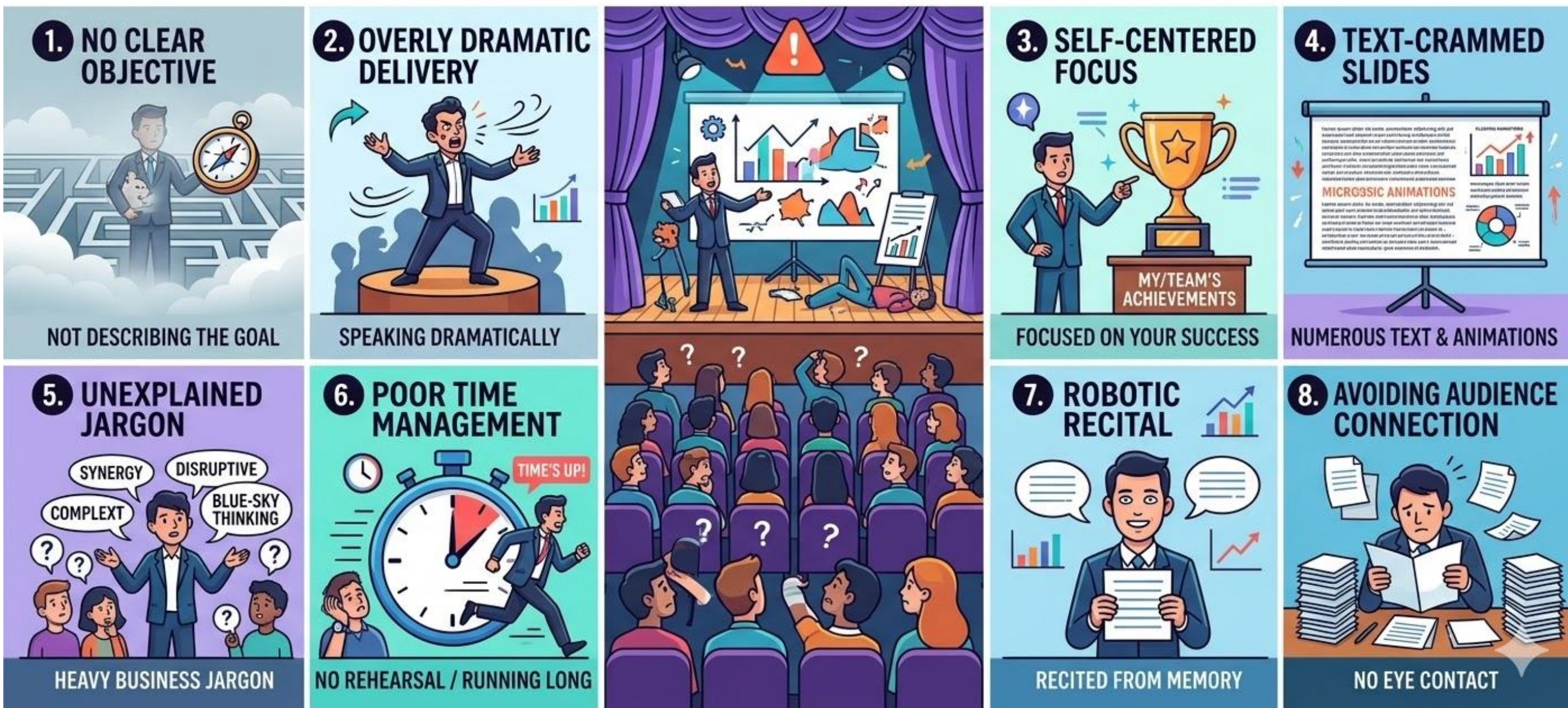
Words, Story, Substance matter - so does delivery!

- Keep it conversational, speak at an appropriate pace
- Make eye contact (do not read the slides)
- Do not keep moving around/fidgeting (sign of nervousness)
- Practice, Practice, Practice!

Ten Ways to Ruin a Pitch

1. Not describing the objective at the beginning
2. Speaking dramatically (instead of conversational)
3. Centered around your/team's achievements
4. Cram your slides with numerous text bullet points and animation
5. Lots of (unexplained) technical/business jargon
6. Not rehearsing to check how long your talk is running
7. Recite from memory
8. Not making eye contact with the audience

8 BIG PITCHING MISTAKES TO AVOID





The Perfect Presentation

Why do some
presentations **change**
the world, others are
forgotten?

3 Steps to Better Presentation Visuals ([link](#))

(Harvard Business Review, 2:35 m)



<https://hbr.org/video/4642346287001/3-steps-to-better-presentation-visuals>

3 Steps to Better Presentation Visuals ([link](#))

(Harvard Business Review, 2:35 m)



Create Slides People Will Remember



How to Give a Killer Presentation ([link](#))

(Harvard Business Review, 3:23 m)



<https://hbr.org/2013/06/how-to-give-a-killer-presentation>

6 Ways to Look More Confident During a Presentation ([link](#))



Steve Jobs Introduced iPhone in 2007 ([link](#))

(YouTube, 10:19 m)

Template: Positioning Statement (Recap)

How to Create a Positioning Statement

For _____
Target Customer

that _____
Needs / Care-Abouts

our product or company is a _____
Category / Solution

that _____
Benefit

Unlike our competitor, our product/company is

Unique Differentiator

Credit: Geoffrey Moore (Crossing the Chasm)

M8 Case Study – Menti Quiz

Menti code: 5127 0708



Essentials of Innovation & Entrepreneurship (EIE) - II

Module (M8): Prototyping

Recall - Pretotype vs. Prototype

Pretotyping

Investment: hours,days

Main Question: Would we use it?

Deliverable: [Working] pretotype



Should We Build It?

Prototyping

Investment: days, weeks

Main Question: Can we build it?

Deliverable: Working prototype



Can We Build It?

Recall - Pretotype vs. Prototype

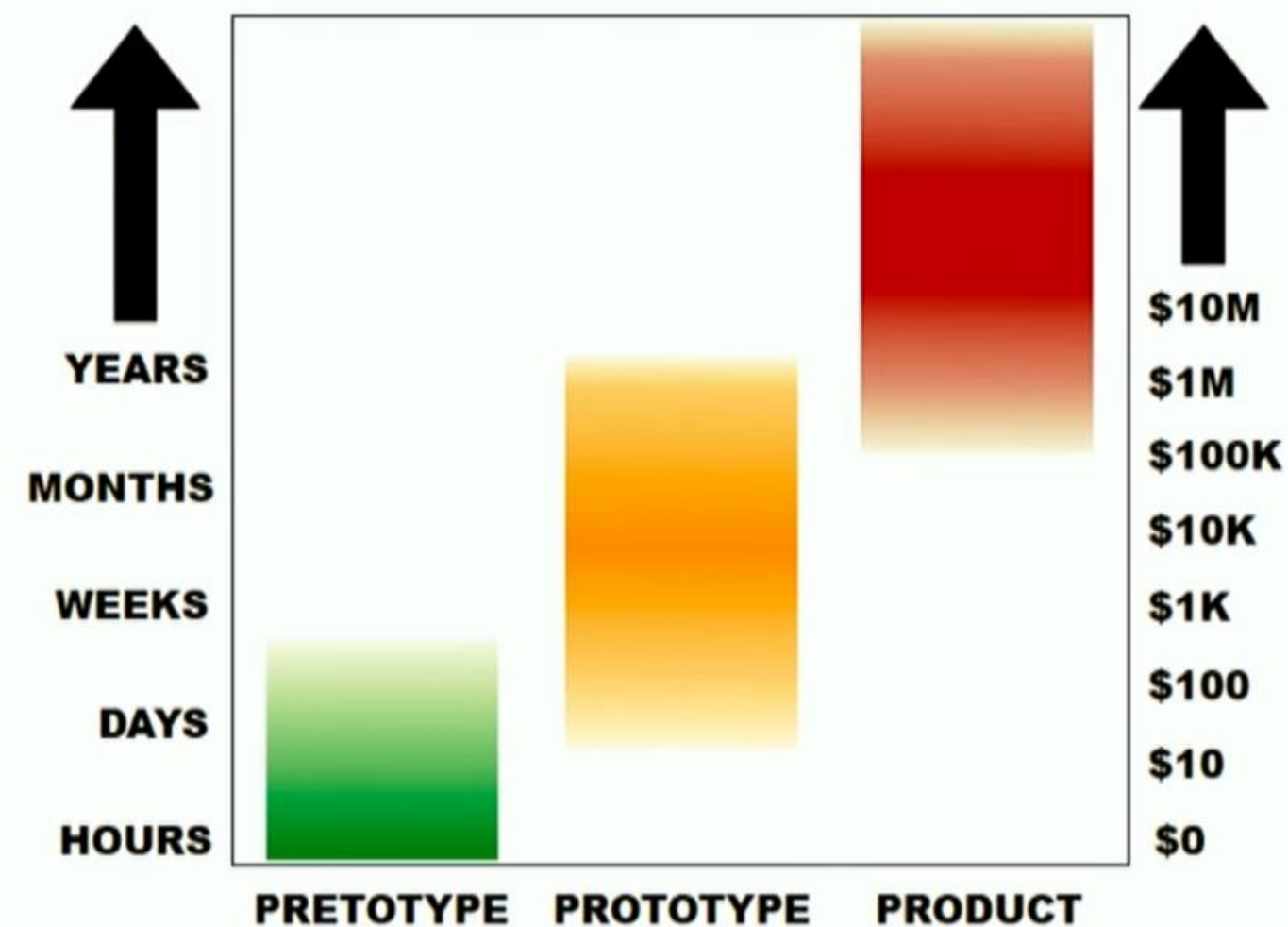
ALBERTO SAVOIA

Stanford Entrepreneurial Thought Leaders Series, March 6, 2019

THE RIGHT IT

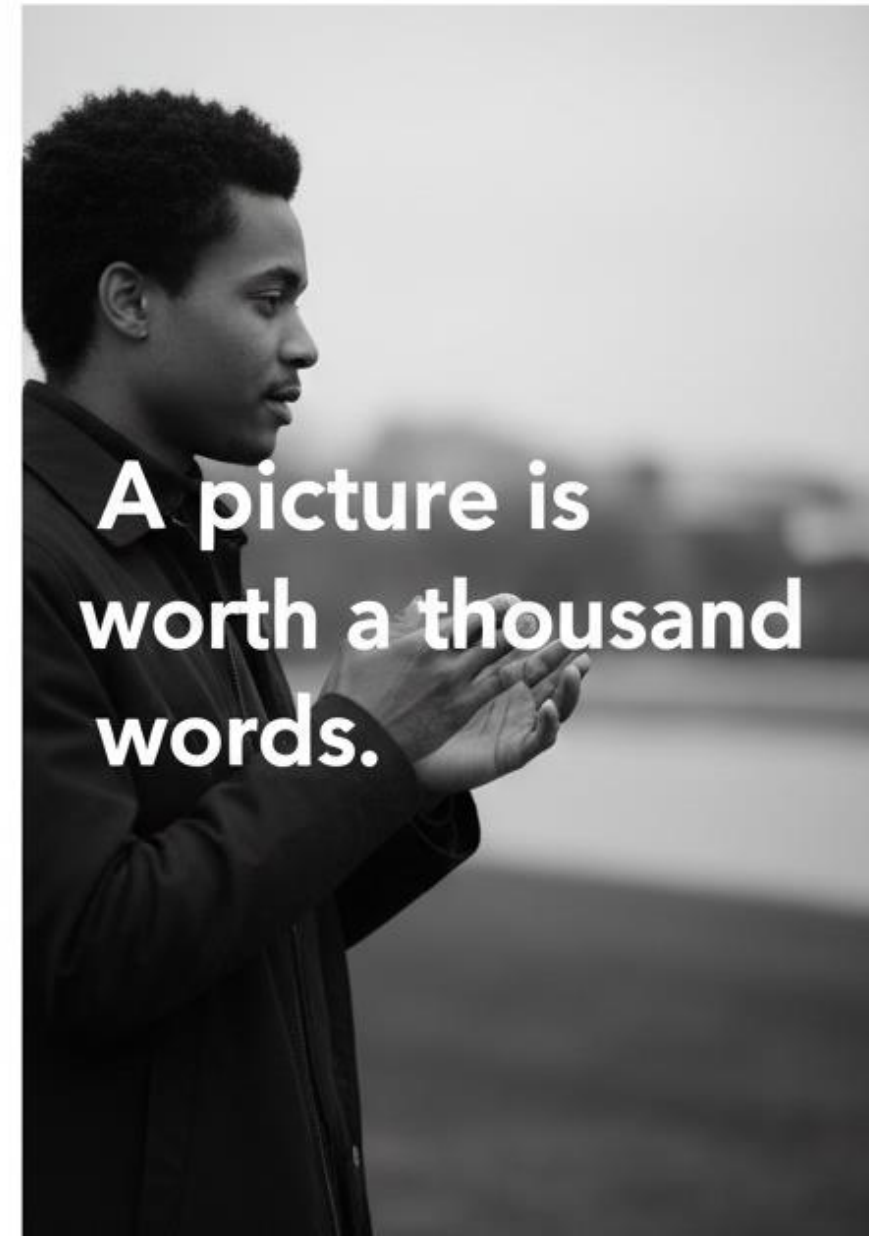
PRETOTYPE:

An artifact or technique that can be used to collect YODA **very quickly** and **very inexpensively**.



Stanford

“



The 1-10-100 Rule: How Early Prototyping Prevents Costly Errors in Advance



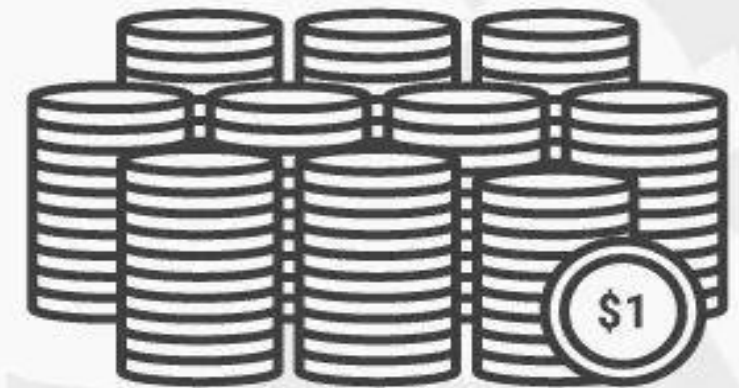
Prevention Cost: \$1

E.g., evaluating usability through early paper prototypes



Correction Cost: \$10

E.g., fixing usability errors discovered through usability tests with hi-fidelity prototypes



Failure Cost: \$100

E.g., fixing the code and lost revenue from an error in the final product

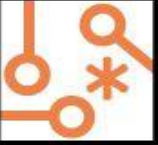
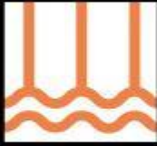


Why Prototype?

- Test ideas and hypotheses related to design and functionality
- Get user feedback – earlier and/or cheaper
- Experiment with alternatives
- Easier to change or throw away
- Reduce planning time & cost by accelerating time-to-market (TTM)
- Learn more about how customers will
(or will not use your product/solution)

Practically, anything can be prototyped (products, services, systems)

Prototype - Terminology

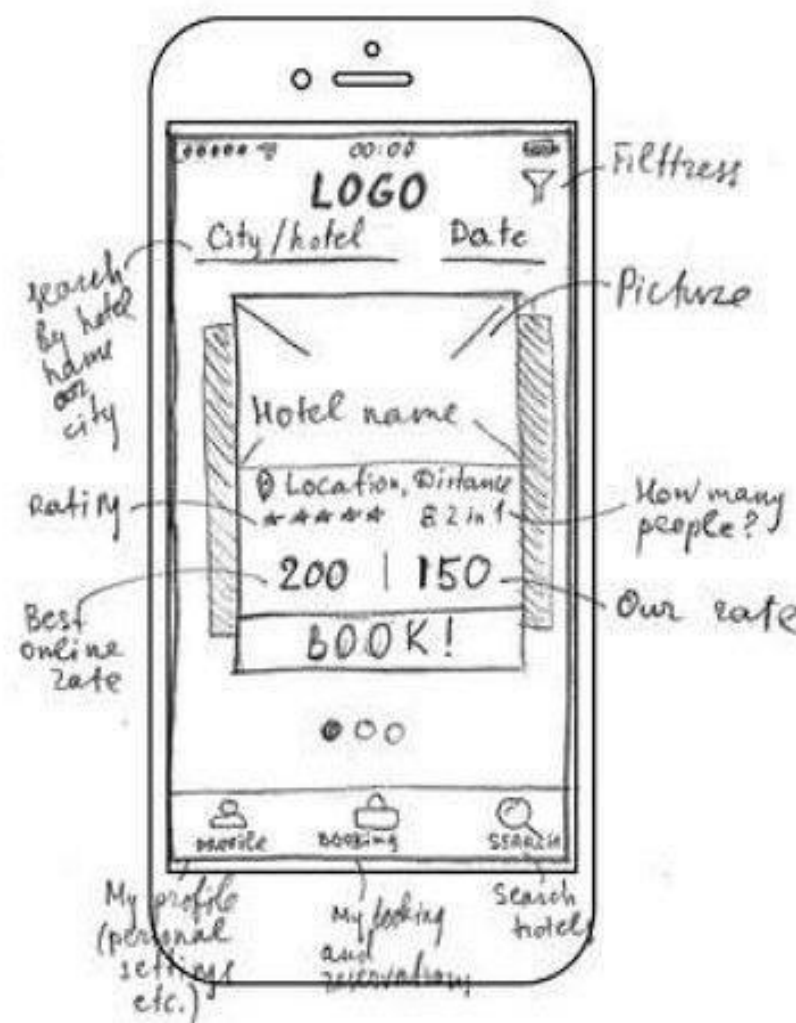
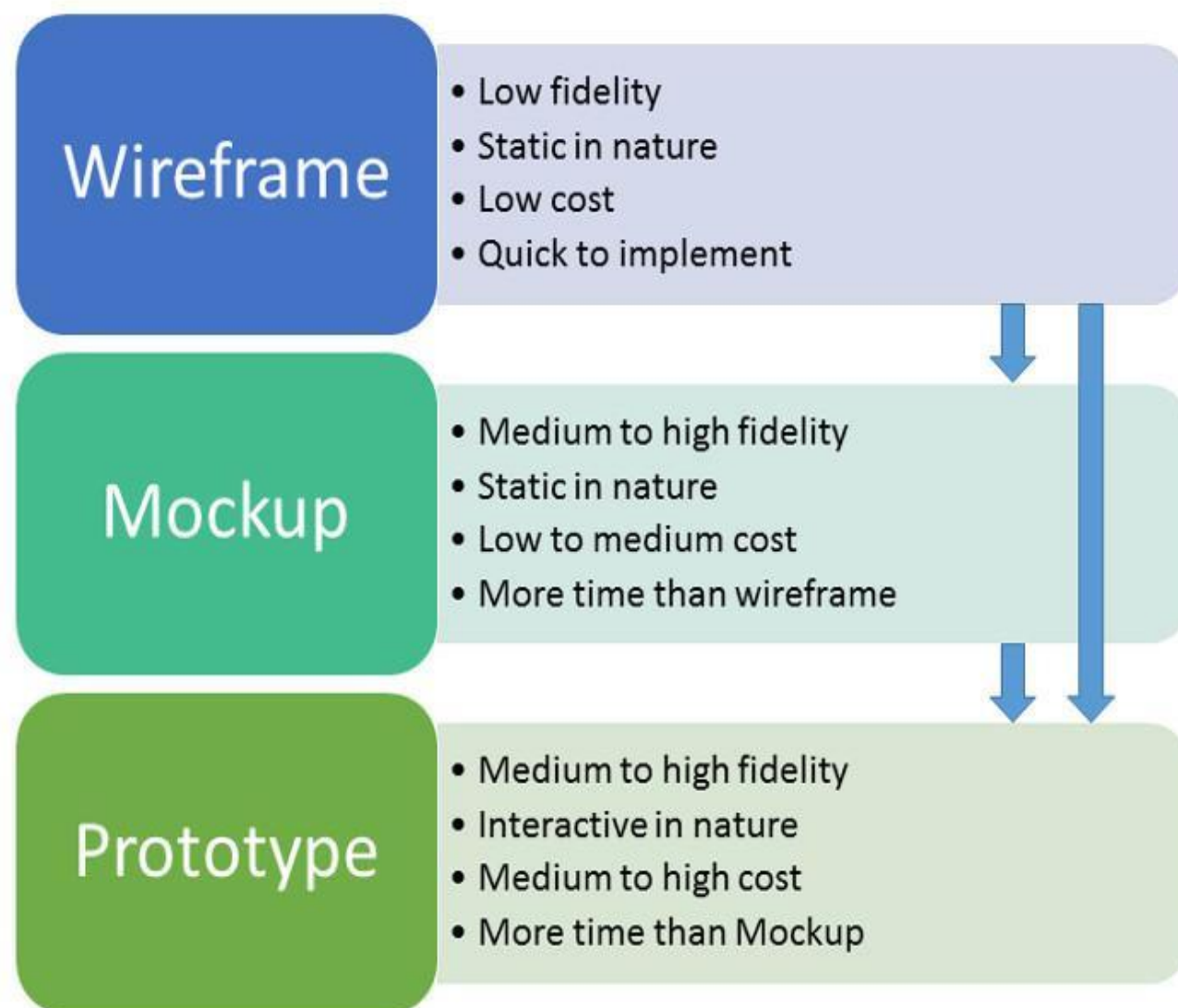
	ALPHA PROTOTYPE: An early prototype, assumed to be unrefined.		BETA PROTOTYPE: A nearly finished, fully functioning version of the product.
	FIDELITY: A prototype's level of sophistication, i.e. low-fidelity or high-fidelity.		RAPID PROTOTYPING: A quick mock-up often employed in interaction design to user-test interactivity and user interfaces. The end result of rapid prototyping is frequently a smoke and mirrors prototype.
	PROOF OF PRINCIPLE: An inexpensively made and unrefined model or prototype; often hacked together using a combination of open-source tools (such as Arduino) and everyday materials (paper, cardboard) to test one aspect of the intended design; also, <i>low-resolution, first-round, first-cut, proof of concept.</i>		SMOKE & MIRRORS, OR WIZARD OF OZ PROTOTYPE: An interactive prototype that works only when someone is behind the scenes, pulling the levers; often employed to user-test interactivity and user interface concepts.

Alpha - Beta

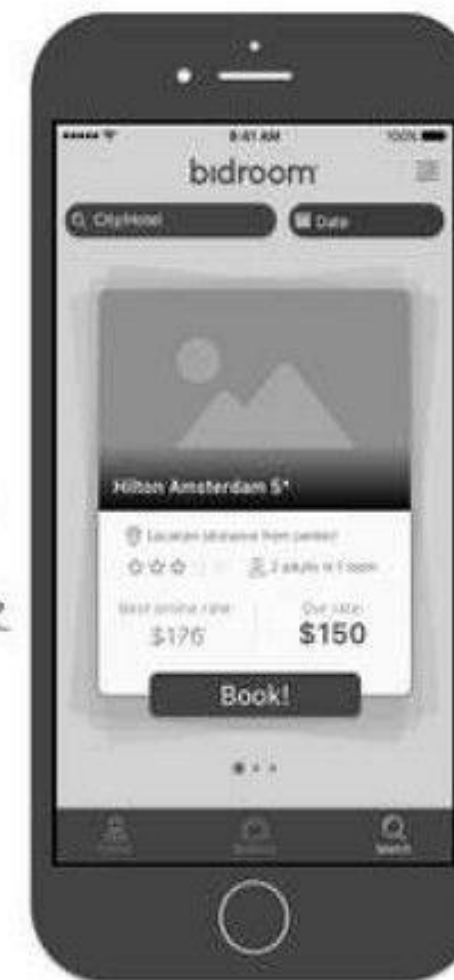


Prototype - Fidelity

Refers to how closely the artifact resembles the final product (degree of details, functionality, etc)



Wireframe

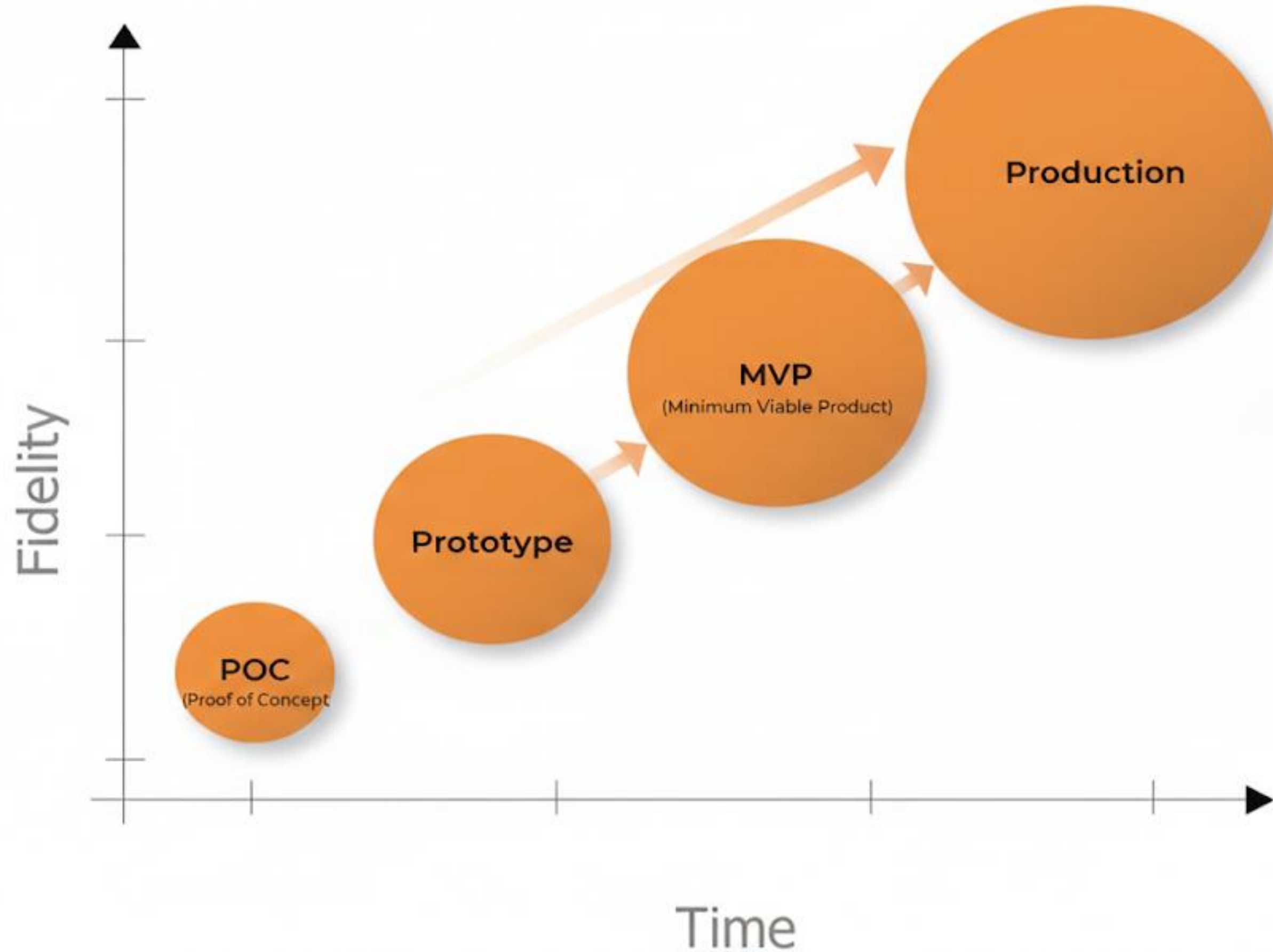


Mockup



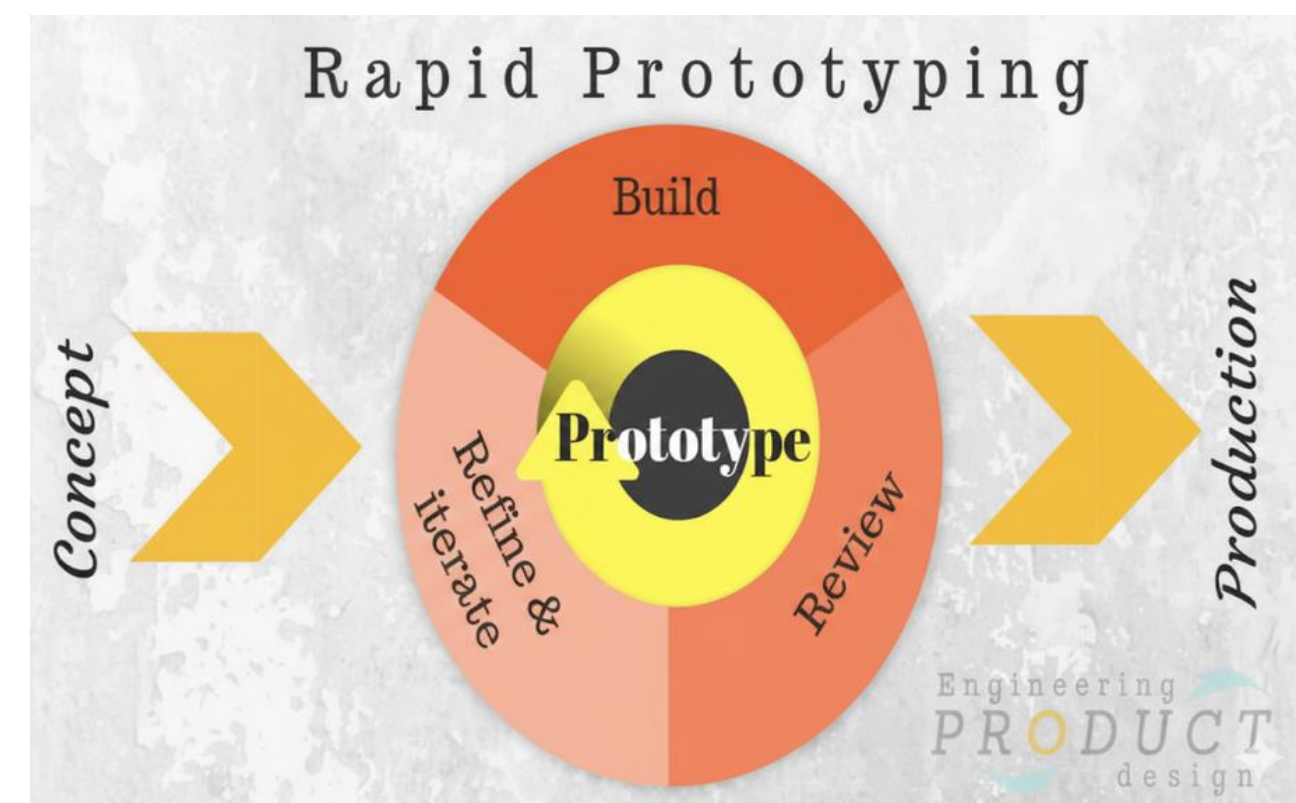
Prototype

Product Development Stages



Rapid Prototyping

- Build what you are testing
- Test it on users, stakeholders or your team
- Adjust/iterate it to reflect feedback
- Repeat the process ('cheaply' in terms of time/money/effort)



Is Prototyping difficult?

- Some years ago, Prototyping used to be constrained by technical skill.
- Today, as technology progresses, the tools have evolved.
- Building something testable is no longer limited to programmers — it is accessible to anyone who can think clearly about a problem.
- **Prototyping Ladder:**
Paper → Digital → LCNC* → Vibe Coding → 3D Print → Functional MVP

** Low Code No Code*

Prototyping in 2026: 'Hold My Coffee'

What is LCNC (Low-Code / No-Code)?

Platforms that allow you to build functional apps using visual interfaces, workflows, and prebuilt components — with minimal traditional coding.

Examples: Bubble, Webflow, Glide etc

What is Vibe Coding?

Building software by describing what you want in natural language and iterating with AI copilots that generate and refine code.

Examples: GitHub Copilot, Replit AI, Cursor, Google Antigravity, Claude Code, Claude Work etc

Breaking the Myth

Myth: "These are only for tech students."

Reality:

- LCNC = Democratized product building
- Vibe coding = AI-assisted logic building
- Both are *idea validation tools*, not "programmer tools"
- Non-tech founders can build MVPs
- Technology still needed for scale, architecture and security considerations

**If you can think clearly about user problems, you can prototype.
Code or Tech is no longer the entry barrier — clarity is.**

Minimum-Viable-Product (MVP)

Before starting with product development, build, the minimum viable product (MVP).

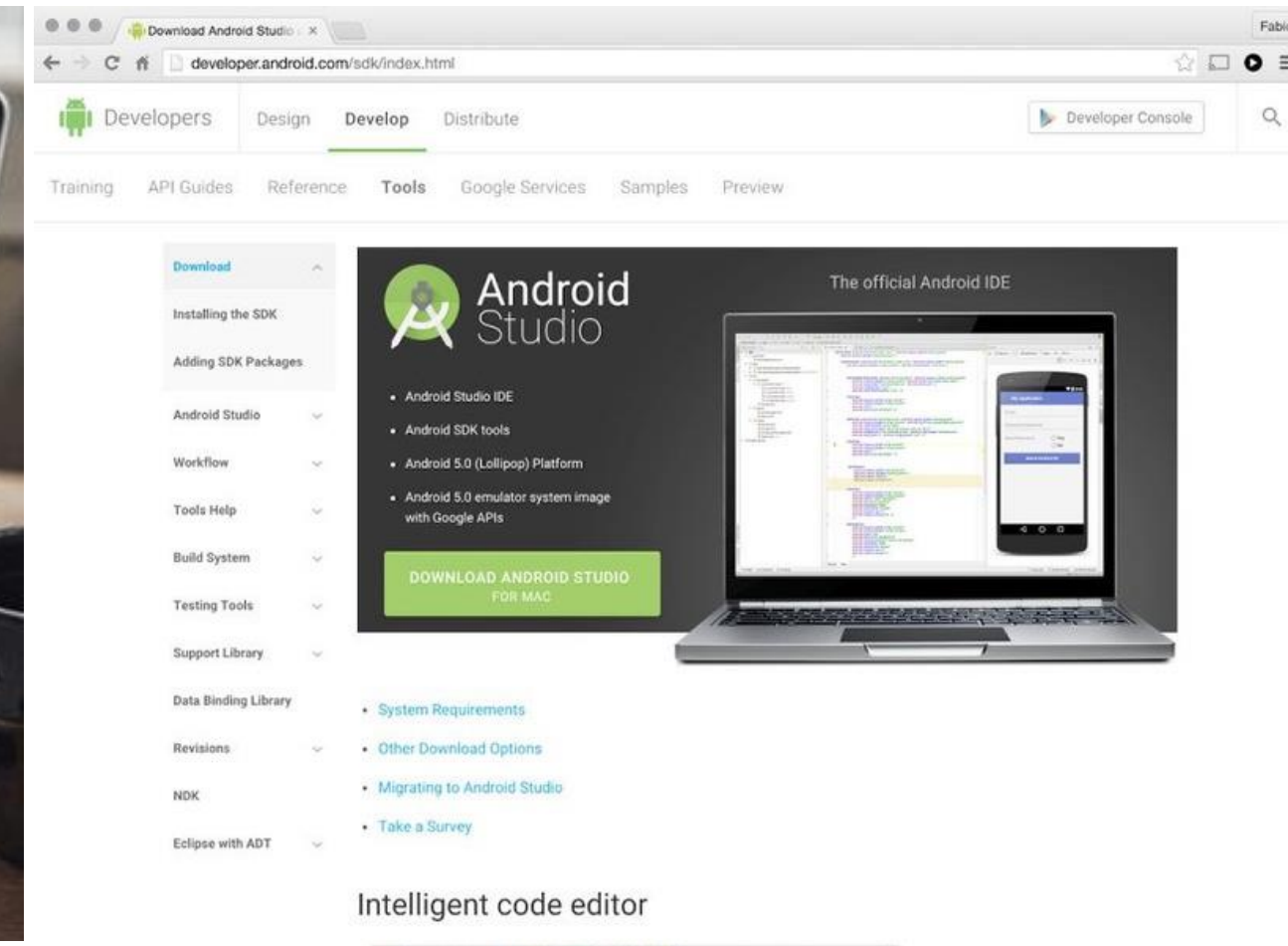
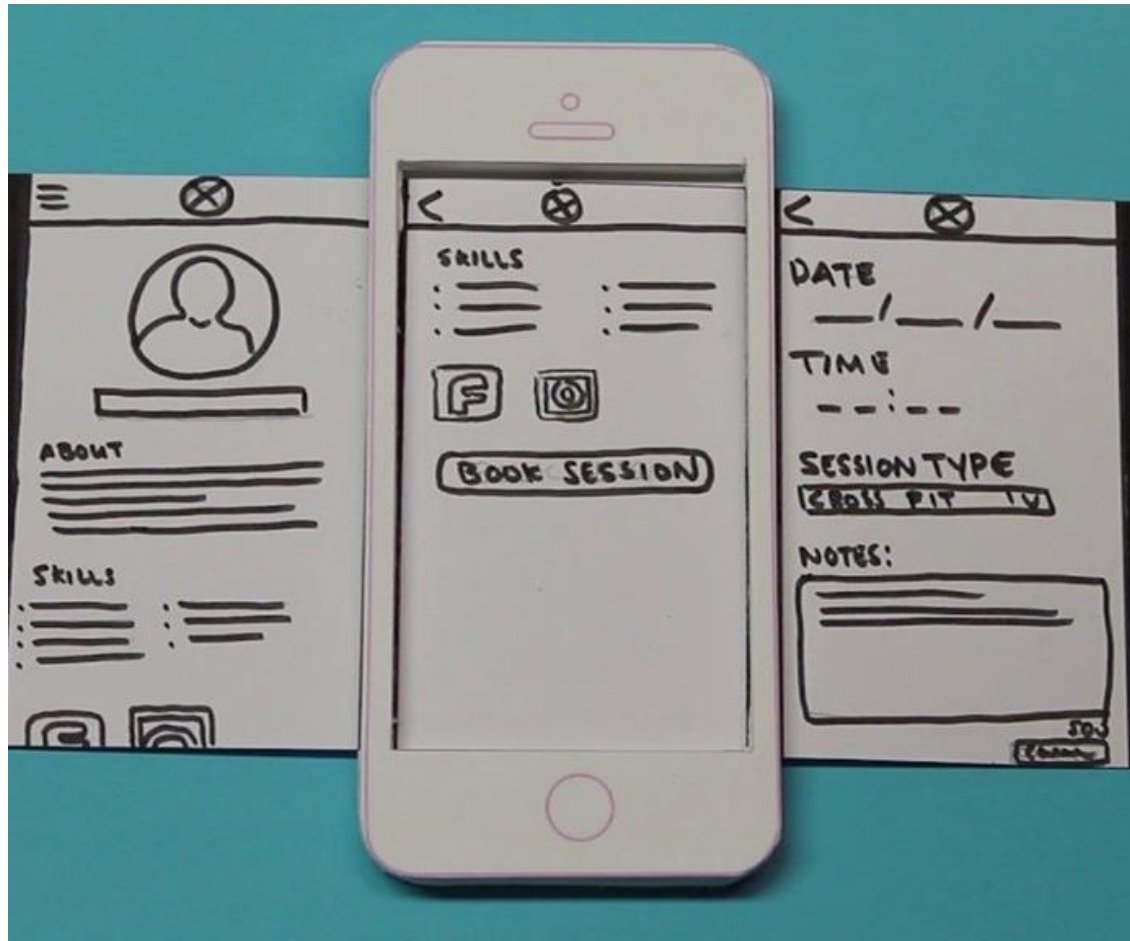
Suggested checklist is on the right (you have to answer 'yes' to all Qs)

MVP CHECKLIST		
Statement	Yes	No
My product - even though with one core feature only in this stage - creates value for customers and thus has the opportunity to get market traction.		
My product has just enough features to test if it is viable in the market.		
I have stripped away everything that is a nice-to-have and kept only the must-have.		
My product's core feature is the one which creates the most value in short term. Other features and updates can be added in a next release.		
This product version allows me the quickest time-to-market while still delivering intended value to customers.		
Even with the minimal feature set, my product can still work correctly and has no bugs.		

Prototyping for Entrepreneurs ([link](#)) (Google)

Sketching and Paper prototyping

Digital Prototyping Native Prototyping



Rapid Prototyping

Prototyping: Idea to Impact



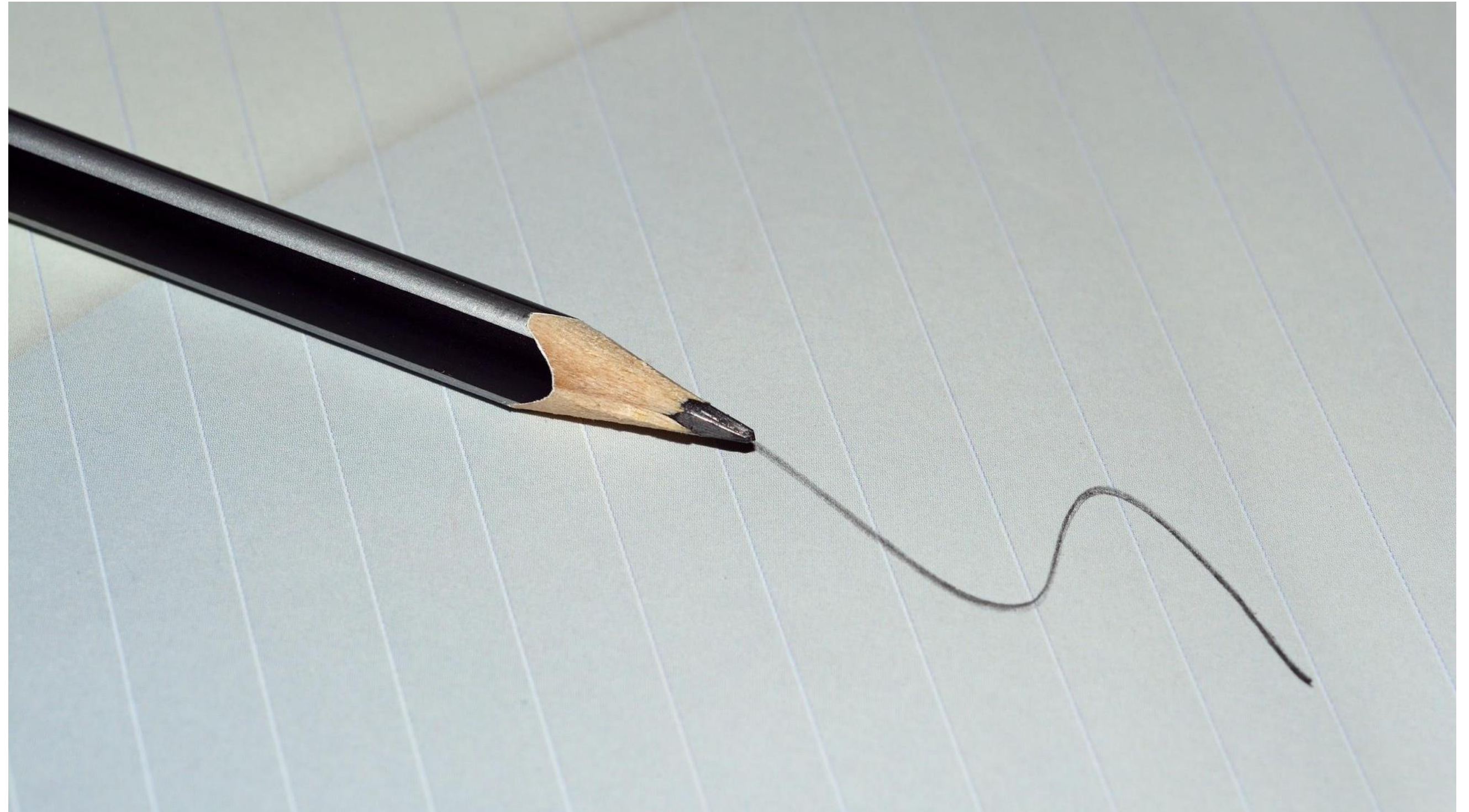
Examples from CIE: Mockup of Carwash in a Container



Prototyping & Testing @ CIE: Team Activity & Fun!



Keep It Simple and Get Started!



For Tomorrow's Innovators

CIE Ignite Season 02: Pitch Week. 82 teams. 499 students.
21 departments. 32 industry mentors. 3 days of live pitching.
From classroom theory to real-world impact!



Pitch Week Status

105/82

Teams Pitching

499

Students

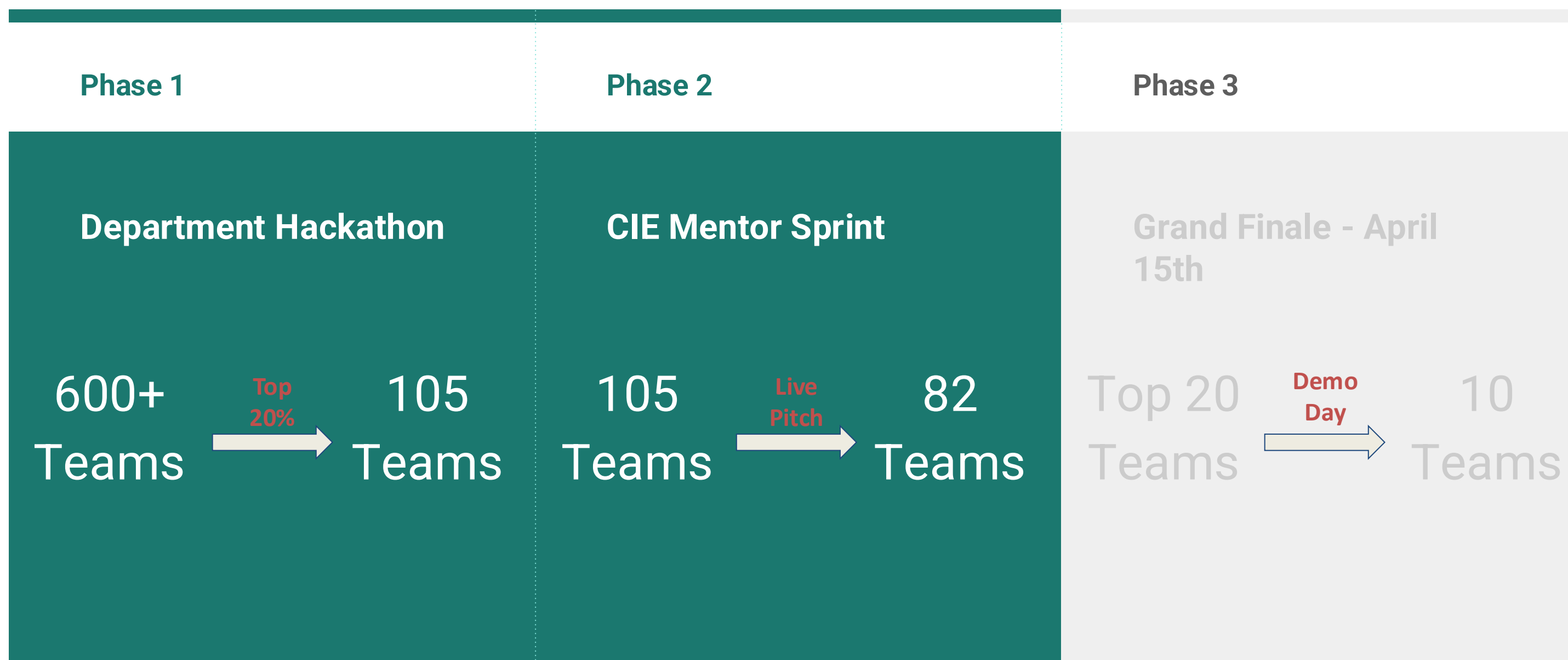
32

Industry Mentors

3

Pitch Days

How it Started



Not a Classroom Exercise.

What 82 teams built and validated in 4 weeks — from soil sensors to CRISPR diagnostics to fall detection wearables.

Customer Validation

-  **1,900+**
customer interviews conducted
-  **64**
teams ran surveys & experiments
-  **23**
Figma / design mockups created
-  **12**
landing pages launched
-  **12**
video demos produced
-  **7**
concierge / manual service tests

What You Built

-  **14**
working prototypes
-  **36**
pretotypes built
-  **29**
validated idea-stage ventures
-  **6**
verticals: SaaS, Hardware, Health...
-  **32**
industry mentors guiding teams
-  **13**
jury panelists evaluating pitches

"1,900+ conversations with users. Not theory. Not assumptions. Real people, real problems."

This Isn't Just an Engineering Program.

21 departments. From Computer Science to Psychology to Law to Design. Everyone has a seat at the table.

CSE 30 teams	ECE 16 teams	AIML 15 teams	BCA 9 teams	BBA 9 teams
Biotechnology 5 teams	B.Des 5 teams	EEE 3 teams	Mechanical 3 teams	Psychology 3 teams
BBA HEM 2 teams	B.Com 2 teams	Economics 1 team	BA LLB 1 team	BBA (General) 1 team

Innovation doesn't belong to one department. It belongs to everyone who shows up.

What's Next?



Phase 3 is Coming - Grand Finale

Top teams from pitch week move forward.
Pre-Incubation. Funding. Demo Day.



Everyone Always has a Chance

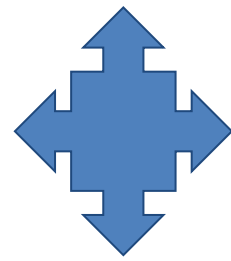
New cohort. New ideas. New teams.
Think your idea is worthy? Come visit us at CIE



82 teams started where you are right now.
A conversation. A problem. A napkin sketch.

Don't watch from the sidelines. Build something.

What's Next?



3D Prototype Workshop @ CIE – April 11

- When: April 11, 10:00 AM – 4:00 PM
- Design and prototype functional, 3D-printable objects
- Begin with a guided beginner build, then explore the CIE to find real problems
- Model your own branded solutions while learning 3D design and print constraints
- Pitch your final model; top 5 designs get printed and displayed in the CIE
- Winner (by participant voting) gets a cash prize, certificate, and feature in the CIE

Limited seats – priority for CIE Ignite teams



Our Website: cie.pes.edu

Reach out: tarunrama@pes.edu

Founder Talk : Murli Thirumale

“Serial Entrepreneur”
**(His First Business Was Acquired By Citrix,
His Second By Dell, And his Third by Pure
Storage after having raised \$50 Million!!)**



Founder Talk • Murli Thirumale

B.Tech IIT BHU • MBA Kellogg School of Management



HP – General Manager



Net6 – Co-founder & CEO



Citrix – SVP & GM



Ocarina Networks - Co-founder and CEO



Dell – Executive



Portworx – Co-founder & CEO (acquired by Pure Storage)



Pure Storage (now Everpure) – GM

3x

Co-founded
companies

25+

Years in
tech & marketing

3

Acquisitions

Expert Talk : Aditya Mehta



Designation:

Co-Founder, Epicure Robotics

Achievements and Recognitions:

Innovator in Robotics, Automation & Mechatronics

- Driven by a passion for solving real-world problems through tech, this founder is revolutionizing food automation as Co-Founder of Epicure Robotics.
- Building automated robotic food kiosks in India to deliver healthy, personalized meals with ultra-convenience.
- For 3 years, they've led mechanical design and robotics efforts at the startup.

Leadership & Technical Contributions:

- IEEE RAS PES University
 - Student Head (2021–2022): Led 35+ teammates and 700+ community members.
- Systemantics India Pvt. Ltd.
 - Embedded Systems Intern (2021–2022): Worked on industrial robotics and embedded tech.

Summary

- “**Persuasion**” is an ART, an Act of reasoning/pleading someone to accept a belief or course of action.
- **Pitching** your Idea/Product/Company is a very important, compelling activity – never to be underestimated.
- Understand – **Ethos**: build Credibility, **Pathos**: create Connection, **Logos**: Prove your Point.
- Focus on your **Audience** – Who, Why and What.
- **Story Telling**, Visual appeal, Effective usage of Data.
- Attention → Evidence → Narrative → Purpose → Impact.
- **Prototype** to get feedback, test hypothesis (design, functionality), Learn about customers, Lesser cost.
- **Understand:**
 - **Pretotypes** prevent you from building the **wrong product**.
 - **Prototypes** prevent you from building the **product wrong**.